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Information Fusion in Financial Markets: A Hierarchical Framework for Combining Social Sentiment, Fundamental Valuation, and Technical Analysis

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Declaration of Authorship

I declare that I have worked on my Master's thesis titled "Information Fusion in Financial Markets: A Hierarchical Framework for Combining Social Sentiment, Fundamental Valuation, and Technical Analysis" by myself and I have used only the sources listed in the bibliography at the end of this thesis.

In Prague, 2026

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Abstract

This thesis investigates whether integrating social media sentiment velocity with earnings announcement timing produces statistically significant abnormal returns in U.S. equity markets. Using a novel dataset of 2,738,767 Reddit posts and comments from fifteen financial subreddits spanning 2010 to 2023, scored using FinBERT for sentiment polarity and matched to 115,518 SEC EDGAR earnings announcement dates for 281 S&P 500 constituents, the study conducts a post-earnings announcement drift (PEAD) event study supplemented by machine learning return prediction. Four hypotheses are tested within a hierarchical signal framework grounded in the investor inattention hypothesis of DellaVigna and Pollet (2009) and Hirshleifer, Lim and Teoh (2009).

H1, that FinBERT sentiment scores predict short-term returns, is not supported in aggregate, though bullish sentiment in the moderate-posts tertile predicts significantly lower twenty-day returns ($t = -2.56$, $p = 0.010$), consistent with noise-trading reversion. H2, that velocity spikes precede abnormal price movements, is supported with strong statistical significance once conditioned on baseline attention level: a 3× velocity spike in high-attention stocks produces a CAR[+1,+5] of −0.61 percent against a no-spike baseline of +0.50 percent ($t = 4.67$, $p < 0.001$), while the effect is entirely absent in zero-attention stocks where velocity ratios are inflated by a denominator floor artefact. H3 confirms that the velocity spike reversal is concentrated in small and high-beta stocks, consistent with the limits-to-arbitrage framework. H4 finds that pre-announcement attention velocity negatively predicts post-announcement drift, consistent with the investor inattention hypothesis: high pre-announcement velocity accelerates information absorption at announcement, compressing subsequent drift. This result is confirmed using an alternative surprise measure and by an independent Google Trends robustness check ($t = 2.44$, $p = 0.015$).

Three machine learning models achieve AUC-ROC of 0.520 on the 2021 to 2023 test period. Feature importance analysis reveals that attention signals (Google Trends 6.5%, Wikipedia 4.8%) outperform raw FinBERT sentiment scores (5.7%) within the social media feature group. A long-short backtest generates a mean return per ML-selected position of 41 basis points and a win rate of 51.5 percent over the test period.

Keywords: information fusion · social media sentiment · Reddit · FinBERT · post-earnings announcement drift · machine learning · attention signals · velocity spikes · event study · U.S. equities

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CHAPTER 1: INTRODUCTION

1.1 Motivation and Research Gap

On the morning of 22 January 2021, a community of retail investors on Reddit's r/WallStreetBets subreddit had been discussing GameStop Corporation, a struggling brick-and-mortar video game retailer, for several days with accelerating intensity. Mention velocity for the ticker GME had exceeded ten times its ninety-day baseline by the time markets opened. What followed over the next six trading days was one of the most dramatic short squeezes in modern financial history: the share price surged from approximately twenty dollars to an intraday peak exceeding four hundred and eighty dollars, inflicting estimated losses of nearly twenty billion dollars on institutional short sellers. The episode was not random noise. It was the product of a measurable, observable, and in retrospect predictable acceleration in social media attention that preceded the price movement by hours, if not days.

This episode was an extreme manifestation of a broader structural transformation in equity markets. The democratisation of trading access through commission-free brokerage platforms, the widespread availability of mobile investment applications, and the emergence of financially oriented social media communities have collectively created a new class of market participant: the retail investor who operates in real time, at scale, and with an unprecedented capacity for collective action. By 2021, retail trading accounted for an estimated twenty-three percent of total U.S. equity market volume, a figure that would have seemed implausible a decade prior (Boehmer, Jones, Zhang and Zhang, 2021). These participants do not process information through the channels assumed by classical asset pricing theory. They form views through social networks, respond to sentiment and attention rather than discounted cash flows, and act on information derived from crowdsourced community analysis rather than professional research.

The academic literature has documented the market impact of social media with growing sophistication. Early work by Antweiler and Frank (2004) demonstrated that internet message board volume predicted stock volatility. Tetlock (2007) showed that media pessimism in the Wall Street Journal predicted downward price pressure followed by mean reversion, a pattern consistent with noise trading rather than fundamental information. Chen, De, Hu and Hwang (2014) established that stock opinions expressed on Seeking Alpha contained incremental information about subsequent returns and earnings surprises, even after controlling for analyst forecasts. More recently, Warkulat and Pelster (2024) documented that attention spikes on r/WallStreetBets reliably preceded uninformed trading behaviour that reduced holding period returns for affected retail investors, with positions opened at peak attention realising average returns of negative 8.5 percent.

Yet several important gaps persist in the existing literature. First, the predominant unit of analysis has been sentiment level, whether the tone of social media posts is positive or negative, rather than sentiment velocity, the rate at which attention and sentiment are accelerating. This distinction matters considerably. A stock that has sustained positive social media sentiment for weeks may already have that sentiment priced in. A stock experiencing a sudden, sharp acceleration in positive mentions (also referred to as a velocity spike) represents a potentially more actionable signal precisely because the information is novel and may not yet be fully incorporated into prices. The January 2021 GameStop episode illustrates this point starkly: the sentiment level on r/WallStreetBets had been constructive for months, but it was the velocity spike, the acceleration to ten times the baseline rate in the days before the price explosion, that marked the actionable event.

Second, existing frameworks typically treat sentiment signals in isolation, without considering how they interact with fundamental information flows. The post-earnings announcement drift literature, one of the most replicated anomalies in finance, documents that stocks with positive earnings surprises continue to outperform in the weeks following announcement, while stocks with negative surprises continue to underperform (Ball and Brown, 1968; Bernard and Thomas, 1989). The investor inattention hypothesis, advanced by DellaVigna and Pollet (2009) and Hirshleifer, Lim and Teoh (2009), proposes that the degree of this drift is moderated by how attentive investors are at the time of the announcement. When fewer investors are engaged, information diffuses more slowly, generating larger and more persistent post-announcement drift. This theoretical lens implies a direct role for pre-announcement social media attention in determining the magnitude of post-announcement drift. No large-scale empirical study has examined this mechanism using a comprehensive Reddit dataset matched to systematic earnings announcement data.

Third, the rapid advancement of financial natural language processing, and specifically the development of transformer-based language models adapted for financial text, has opened new possibilities for sentiment measurement that have not yet been fully exploited in the post-earnings announcement drift literature. The FinBERT model developed by Huang, Wang and Yang (2023) represents a substantial methodological advance, demonstrating superior performance on financial sentiment classification tasks compared to naïve Bayes, support vector machines, random forests, and general-purpose BERT models, precisely because it leverages a large corpus of domain-specific financial text during pre-training.

This thesis investigates whether integrating Reddit-derived sentiment velocity signals with earnings announcement timing produces statistically significant abnormal returns in U.S. equity markets. The empirical dataset comprises 2,738,767 Reddit posts and comments from fifteen financial subreddits spanning January 2010 through December 2023, fully scored using FinBERT for sentiment polarity and matched to 115,518 earnings announcement dates from SEC EDGAR for 281 S&P 500 constituent firms. This is one of the most comprehensive such datasets assembled for this purpose. The primary analytical tool is an event study measuring cumulative abnormal returns around earnings announcement dates, supplemented by machine learning return prediction models following Gu, Kelly and Xiu (2020) to assess the incremental predictive content of social sentiment signals across the cross-section of U.S. equities.

1.2 Research Objectives

This thesis pursues four interconnected research objectives, each motivated by the gaps identified above and each contributing to the broader goal of understanding how social media attention interacts with investor inattention and earnings catalysts in the price discovery process for U.S. equities.

The first objective is to construct and document a novel, large-scale dataset integrating historical Reddit sentiment with earnings announcement dates and equity price data. This requires collecting and processing 2,738,767 Reddit mentions from fifteen financial subreddits spanning 2010 to 2023, applying FinBERT sentiment scoring to each mention to extract positive, negative, and neutral probability scores, and matching this data to SEC EDGAR 8-K, 10-Q, and 10-K filing dates and daily equity price data from Stooq.

The second objective is to empirically test four hypotheses about the relationship between Reddit signals and equity returns. H1 tests whether FinBERT sentiment scores predict short-term returns. H2 tests whether attention velocity spikes precede abnormal price movements. H3 examines whether these effects vary systematically across stock characteristics. H4 tests whether pre-announcement attention velocity modifies post-earnings drift through the investor inattention channel. All four hypotheses are evaluated using an event study framework that measures cumulative abnormal returns relative to a market model benchmark across pre-announcement, announcement, and post-announcement windows.

The third objective is to apply machine learning methods, following the approach of Gu, Kelly and Xiu (2020), to determine the optimal combination of sentiment, attention, and fundamental signals for predicting equity returns out of sample. Three models, logistic regression, random forest, and gradient boosting, are trained on data from 2010 to 2020 and evaluated on the held-out test period from 2021 to 2023, with feature importance analysis revealing which signal layers contribute most to predictive accuracy and a long-short backtest providing an assessment of economic significance.

The fourth objective is to situate the findings within the broader theoretical literature on market efficiency, investor inattention, behavioural finance, and machine learning in asset pricing, identifying the specific contribution of this research relative to existing work and discussing the practical implications for quantitative investors and market structure researchers.

1.3 Research Questions and Hypotheses

The primary research question guiding this thesis is: does integrating social media sentiment velocity with earnings announcement timing produce statistically significant abnormal returns in U.S. equity markets, and can machine learning methods determine optimal signal weights across stock characteristics?

Four testable hypotheses derive from the theoretical framework. Hypothesis 1 (H1): Social media sentiment signals extracted from Reddit financial communities using FinBERT demonstrate statistically significant predictive power for short-term equity returns. Hypothesis 2 (H2): Mention velocity spikes, defined as instances where the seven-day rolling mention rate exceeds the ninety-day baseline by a factor of three, five, or ten, precede abnormal price movements within a 24 to 48-hour window, with the effect concentrated in stocks with high baseline Reddit attention. Hypothesis 3 (H3): The predictive power of social sentiment and attention velocity varies systematically with stock characteristics, with stronger effects observed in retail-dominated, smaller-capitalisation, and higher-volatility securities. Hypothesis 4 (H4): Pre-earnings announcement attention velocity modifies post-earnings announcement drift in a manner consistent with the investor inattention hypothesis, with stocks exhibiting low pre-announcement velocity predicted to show larger subsequent drift.

1.4 Scope and Delimitations

The asset universe is restricted to U.S. equities, specifically historical constituents of the S&P 500 index including delisted firms, spanning 2010 to 2023. The primary social sentiment source is Reddit. Google Trends search interest data and Wikipedia page view data serve as supplementary attention proxies and are used specifically as robustness checks for the H2 and H4 velocity analyses. The FinBERT model (ProsusAI/finbert) is used for all sentiment scoring. The study period spans January 2010 through December 2023, with machine learning models trained on 2010 to 2020 data and evaluated on the 2021 to 2023 test period. No live trading strategy is implemented with fully specified transaction costs and capital allocation rules; the backtest reported in Section 5.6 is an event-level assessment intended to provide a first-order assessment of economic significance rather than a deployable investment strategy.

1.5 Structure of the Thesis

Chapter 2 reviews the academic literature across five principal areas: the efficient market hypothesis and its documented violations; the investor inattention hypothesis and its implications for post-earnings announcement drift; the growing empirical literature on social media, investor attention, and asset prices; machine learning approaches to return prediction; and financial natural language processing. Chapter 3 develops the theoretical framework, specifying the hierarchical signal architecture and deriving the four hypotheses from first principles, with particular attention to the inattention mechanism as the theoretical basis for the H4 prediction. Chapter 4 describes the data construction process in detail and addresses data quality considerations including survivorship bias and the event date definition challenge. Chapter 5 presents the empirical results, progressing from descriptive statistics through the four hypothesis tests before presenting the machine learning results, feature importance analysis, and backtest performance. Chapter 6 discusses the findings in relation to the existing literature and identifies practical implications. Chapter 7 concludes with a summary of contributions and directions for future research.

CHAPTER 2: LITERATURE REVIEW

This chapter reviews the academic foundations underlying the thesis across five principal areas. Section 2.1 examines the theoretical foundations of market efficiency and its behavioural finance critiques, including the investor inattention hypothesis and its implications for post-earnings announcement drift. Section 2.2 surveys the empirical literature on social media, investor attention, and asset prices. Section 2.3 reviews the machine learning asset pricing literature. Section 2.4 discusses financial natural language processing and FinBERT specifically. Section 2.5 synthesises the gaps across these streams to motivate the specific contribution of this thesis.

2.1 Market Efficiency, Behavioural Finance, and the PEAD Anomaly

2.1.1 The Efficient Market Hypothesis and Its Limits

The efficient market hypothesis (EMH), articulated in its canonical form by Fama (1970), asserts that financial asset prices fully and instantaneously incorporate all available information. Under the semi-strong form, which is the variant most directly relevant to this thesis, prices reflect all publicly available information including historical prices, accounting data, and published news. Any attempt to earn abnormal returns by trading on publicly available information should therefore be futile in competitive equilibrium.

A substantial body of anomaly literature accumulated from the 1980s onward documenting systematic patterns in returns that are difficult to reconcile with strict efficiency. These anomalies include the size effect (Banz, 1981), the value premium (Fama and French, 1992), short-term momentum in returns (Jegadeesh and Titman, 1993), and most pertinently for this thesis, post-earnings announcement drift (Ball and Brown, 1968; Bernard and Thomas, 1989). Lo (2004) proposed the Adaptive Markets Hypothesis as a more nuanced framework that reconciles the EMH with behavioural finance observations, predicting that attention-driven mispricings can persist for exploitable periods when arbitrage faces real constraints. The limits-to-arbitrage literature formalises these constraints (Shleifer and Vishny, 1997). Noise trader risk, in the language of De Long, Shleifer, Summers and Waldmann (1990), imposes a constraint on rational arbitrage that allows sentiment-driven deviations from fundamental value to persist.

2.1.2 Investor Sentiment and the Cross-Section of Returns

Baker and Wurgler (2006, 2007) provided the most comprehensive empirical treatment of investor sentiment effects on the cross-section of stock returns. Their 2006 Journal of Finance study constructs a composite sentiment index from six proxies and documents that beginning-of-period sentiment strongly predicts the subsequent cross-section of returns, with effects concentrated in stocks that are difficult to value and costly to arbitrage. When sentiment is high, small stocks, young stocks, high-volatility stocks, and distressed stocks earn relatively low subsequent returns as mispricing reverses. This framework motivates Hypothesis 3 of this thesis, which predicts that effects of Reddit-derived signals are stronger for smaller-capitalisation, higher-volatility, retail-dominated securities. The fifteen most-mentioned tickers in the dataset, including GameStop (161,489 mentions), AMC Entertainment (62,478), and Palantir (13,007), are precisely the type of securities that Baker and Wurgler identify as most vulnerable to sentiment effects. Barber and Odean (2008) provide a complementary behavioural mechanism: retail investors disproportionately purchase stocks that recently attracted unusual attention, regardless of whether that attention reflects positive or negative news.

2.1.3 Post-Earnings Announcement Drift

Post-earnings announcement drift (PEAD) is the tendency for stock prices to continue drifting in the direction of an earnings surprise for weeks following the announcement. Ball and Brown (1968) first identified the pattern, noting that prices did not instantaneously and completely adjust to the information in earnings reports. Bernard and Thomas (1989) provided the definitive characterisation of the anomaly, documenting that a zero-investment strategy of buying positive-surprise stocks and short-selling negative-surprise stocks earned economically and statistically significant abnormal returns over the sixty trading days following announcement. The persistence of PEAD over decades of academic scrutiny is theoretically puzzling. Jensen, Kelly and Pedersen (2023) document that many classical anomalies, including earnings-based strategies, have weakened substantially in the post-2000 period as the academic literature drew attention to them. For this thesis, PEAD serves as both an empirical anchor and a theoretical mechanism: Hypothesis 4 proposes that pre-announcement social media attention velocity modifies the magnitude of PEAD through the investor inattention channel.

2.1.4 Investor Inattention and Post-Earnings Announcement Drift

A distinct and empirically well-supported explanation for PEAD centres on investor inattention. The core insight is that investors have finite cognitive capacity to process information, and the completeness of price adjustment depends on how attentive the relevant investor population is at the time of the announcement. When investors are distracted or disengaged, information is absorbed more slowly, the initial price response is incomplete, and subsequent drift is larger as the information gradually diffuses through the market.

DellaVigna and Pollet (2009) provide direct evidence for this mechanism by exploiting exogenous variation in investor attention. They document that earnings announcements made on Fridays, when investors are less attentive than on other weekdays, produce significantly larger post-announcement drift. The Day-0 price response to Friday announcements is smaller and subsequent drift is larger, consistent with incomplete information processing at announcement followed by delayed correction. Hirshleifer, Lim and Teoh (2009) extend this analysis by documenting that PEAD is significantly stronger on days when many firms report earnings simultaneously, creating an information overload environment. Hou, Peng and Xiong (2009) examine the relationship between investor attention and the momentum anomaly, finding that stocks with lower investor attention exhibit stronger momentum effects, supporting the broader principle that inattention-driven underreaction generates return predictability.

Together, these studies establish a theoretical framework directly applicable to the Reddit velocity analysis in this thesis. Pre-announcement Reddit velocity is a proxy for the retail investor attention environment at the time of an earnings announcement. Low pre-announcement velocity indicates a less attentive retail investor base, predicting slower information absorption and larger post-announcement drift. High pre-announcement velocity indicates a more attentive retail investor base, predicting more complete initial price adjustment and smaller subsequent drift. This is the theoretical basis for Hypothesis 4.

2.2 Social Media, Investor Attention, and Asset Prices

2.2.1 Early Internet Sentiment Research

Antweiler and Frank (2004) examined the relationship between message volume and content on Yahoo Finance message boards, finding that message volume predicted subsequent trading volume and that bullish postings were associated with higher next-day volatility. Tetlock's (2007) landmark Journal of Finance study quantified the relationship between media pessimism and market outcomes, documenting that high pessimism robustly predicted downward pressure on market prices followed by mean reversion. The pattern of price impact followed by reversal is characteristic of noise trading in the De Long et al. (1990) framework. Chen, De, Hu and Hwang (2014) examined the value of stock opinions on Seeking Alpha, documenting that the aggregate sentiment in 97,000+ articles predicted subsequent stock returns and earnings surprises, with the effect concentrated in stocks with low institutional ownership and high information asymmetry.

2.2.2 Reddit and the WallStreetBets Literature

The emergence of Reddit's r/WallStreetBets subreddit as a major force in financial markets generated substantial new literature. Betzer and Harries (2022) documented a significant and positive relationship between Reddit comment volume and subsequent trading measures, consistent with an attention-based mechanism. Notably, they found no evidence that Reddit posts contained fundamental information, suggesting the attention effect was informationally empty but nonetheless sufficient to move prices in a market with high retail participation. Warkulat and Pelster (2024) found that WSB attention significantly reduces holding period returns, with positions created when WSB attention is at its highest realising negative 8.5 percent holding period returns on average, identifying Reddit-driven trading as predominantly uninformed. Bradley, Hanousek, Jame and Xiao (2024), published in the Review of Financial Studies, found that WSB posts predict short-term stock returns consistent with price pressure from retail demand, concentrated in stocks with high short interest and retail ownership.

2.2.3 Attention Proxies: Search Volume and Wikipedia Page Views

Da, Engelberg and Gao (2011) established the Abnormal Search Volume Index, constructed from Google Trends data, as a reliable proxy for retail investor attention. Stocks in the top quintile of weekly abnormal search volume subsequently experienced a price run-up followed by reversal, consistent with the attention-driven buying mechanism of Barber and Odean (2008). Ben-Rephael, Da and Israelsen (2017) found that institutional and retail attention have different informational content: retail attention is more associated with subsequent return reversal, consistent with the noise trading interpretation. Moat et al. (2013) documented that increases in Wikipedia page views for financial topics preceded stock market moves, suggesting that information-seeking behaviour is a leading indicator of investor attention. This thesis employs both Google Trends and Wikipedia page views as supplementary attention proxies in the robustness checks for H2 and H4.

2.3 Machine Learning in Asset Pricing

2.3.1 Gu, Kelly and Xiu (2020)

Gu, Kelly and Xiu (2020), published in the Review of Financial Studies, provides the primary methodological framework for the signal combination approach employed in this thesis. The paper examines nearly one hundred stock characteristics, applying machine learning methods to the problem of predicting monthly stock returns out of sample. Their principal finding is that machine learning methods substantially outperform ordinary least squares regression in out-of-sample return prediction, with the improvement concentrated in nonlinear interaction effects among predictors that linear models cannot capture. This thesis follows their temporal train-test split philosophy, training all models on 2010 to 2020 data and evaluating on the held-out 2021 to 2023 period. Giglio, Kelly and Xiu (2022), in their Annual Review of Financial Economics survey, situate the machine learning asset pricing literature within the broader context of factor model research, motivating the inclusion of traditional controls alongside sentiment features in the prediction models.

2.3.2 Murray, Xia and Xiao (2024) and Brogaard and Zareei (2023)

Murray, Xia and Xiao (2024), published in the Journal of Financial Economics, demonstrate that machine learning methods can identify return-predictive price chart patterns that differ substantially from the manually defined patterns in traditional technical analysis. Their finding that learned patterns exhibit context independence supports the generalisability of machine learning approaches to the return prediction problem studied here. Brogaard and Zareei (2023), published in the Journal of Financial and Quantitative Analysis, find that gradient boosting classifiers, the same model class employed in this thesis, generate Sharpe ratios of approximately 1.0 on long-short strategies, with the performance robust to transaction cost adjustments. Critically, they document that predictive performance is substantially higher when alternative data signals including sentiment-based measures are included alongside traditional predictors, directly motivating the multi-source signal architecture of this thesis.

2.3.3 Ke, Kelly and Xiu (2021)

Ke, Kelly and Xiu (2021) introduce a supervised text mining methodology that extracts return-predictive signals from news articles by optimising text-derived scores specifically for the prediction target of stock returns. They find that information is assimilated into prices with an inefficient delay concentrated in smaller, more volatile stocks, consistent with the limits-to-arbitrage framework. Their finding that the efficiency delay varies systematically with arbitrage costs motivates Hypothesis 3, which predicts that Reddit-derived sentiment and attention signals will exhibit stronger predictive power for retail-dominated, higher-volatility securities where the conditions for slow information diffusion are most pronounced.

2.4 Financial Natural Language Processing and FinBERT

Sentiment analysis of financial text has evolved through three generations of methodology. The first generation, exemplified by the Loughran and McDonald (2011) financial sentiment dictionary, applied word count-based approaches adapted to the domain-specific vocabulary of financial text. Dictionary-based approaches are computationally efficient and interpretable, but they fail to capture contextual meaning, irony, and the complex sentence structures characteristic of informal social media text.

The third generation, represented by transformer-based language models adapted for financial text, overcomes the limitations of both earlier approaches. Huang, Wang and Yang

(2023) developed FinBERT, a version of Google's BERT model pre-trained on a corpus of 4.9 billion tokens from financial text and fine-tuned for sentiment classification on a labelled dataset of 10,000 analyst report sentences. FinBERT achieved substantially higher accuracy than the Loughran-McDonald dictionary, naive Bayes, support vector machines, random forests, CNN, and LSTM models, and outperformed general-purpose BERT. Kirtac and Germano (2024) subsequently evaluated FinBERT on nearly one million U.S. financial news articles from 2010 to 2023, finding that while larger language models slightly outperformed FinBERT in direct return prediction accuracy, FinBERT provided strong second-best performance and was considerably more computationally tractable at scale. The ProsusAI/finbert implementation used in this thesis classified 2,738,767 Reddit mentions with a distribution of neutral 83.6 percent, negative 8.9 percent, and positive 7.4 percent. The application of FinBERT to Reddit text involves a known domain mismatch: the model was trained primarily on formal financial documents, while Reddit posts exhibit informal language, abbreviations, memes, and sarcasm. This mismatch likely attenuates the estimated sentiment-return relationship toward zero, making the results conservative.

2.5 Research Gap and Contribution

The five streams of literature reviewed above have each advanced substantially in recent years. Yet their intersection remains underdeveloped. No study has combined a large-scale historical Reddit dataset, FinBERT sentiment scoring, EDGAR earnings announcement dates, the investor inattention hypothesis, and machine learning return prediction within a unified empirical framework designed to test whether pre-announcement social media attention velocity modifies post-earnings drift.

The specific contributions of this thesis are threefold. First, it constructs a novel dataset comprising 2,738,767 Reddit mentions from fifteen subreddits spanning 2010 to 2023, fully scored using FinBERT, matched to 115,518 earnings announcement dates for 281 S&P 500 constituent firms, and supplemented by 820,302 Wikipedia page view observations and 158,542 Google Trends data points. Second, this thesis introduces and empirically tests a bivariate framework for attention signal construction that conditions velocity spikes on baseline attention level, resolving an important methodological ambiguity in the existing velocity spike literature. Third, this thesis provides the first large-scale empirical test of the investor inattention hypothesis in a social media context, using pre-announcement Reddit velocity as a direct proxy for the retail investor attention environment at the time of an earnings announcement.

CHAPTER 3: THEORETICAL FRAMEWORK

This chapter develops the conceptual architecture that motivates the empirical design. Section 3.1 develops a stylised model of information flow in equity markets. Section 3.2 specifies the three-layer hierarchical signal architecture. Section 3.3 presents the PEAD-attention interaction model. Section 3.4 explains why machine learning is the appropriate tool for combining signals across layers. Section 3.5 maps each of the four hypotheses to a specific theoretical prediction.

3.1 Information Flow in Equity Markets

Equity prices aggregate information from heterogeneous sources arriving at different frequencies, with different degrees of precision, and processed by different investor populations. The empirical evidence reviewed in Chapter 2 presents a more textured picture than the EMH predicts: different types of information are incorporated at different speeds, and the degree of underreaction is systematically related to the investor composition of the stock and the nature of the information source.

A stylised model of information flow identifies three broad categories of signal with distinct temporal footprints. Fundamental information, including earnings surprises, guidance revisions, and regulatory filings, arrives discretely at scheduled announcement dates and has a long half-life, as evidenced by PEAD. Technical information, including price momentum, volume patterns, and moving average signals, is continuously available at daily frequencies. Social sentiment and attention signals occupy a distinct temporal niche: they are generated continuously at near-real-time frequencies and reflect the instantaneous attention and interpretation of the retail investor community.

The critical distinction motivating this framework is between the level and the velocity of social attention. A sustained sentiment signal at a constant level may already be fully priced. An acceleration in attention, by contrast, signals the emergence of new collective interest that has not yet been fully acted upon. A second critical distinction is between the attention level itself and the spike relative to that level: a tenfold increase in mentions from a near-zero baseline is fundamentally different from a tenfold increase for a stock that is already actively discussed. The bivariate signal architecture formalises this distinction.

3.2 The Hierarchical Signal Architecture

3.2.1 Layer 1: Valuation Anchor (Low Frequency)

The first layer establishes the fundamental valuation context at quarterly frequency. In the empirical implementation of this thesis, the primary fundamental signal is the Day-0 abnormal return at earnings announcement, which serves as a proxy for the Standardised Unexpected Earnings measure used in the PEAD literature. Within the hierarchical framework, Layer 1 establishes the directional prior for post-announcement drift and defines the benchmark against which the incremental contribution of Layer 2 signals is assessed.

3.2.2 Layer 2: Sentiment and Attention (Medium Frequency)

The second layer operates at daily frequency and constitutes the empirically novel core of the framework. It comprises three distinct sub-signals: FinBERT sentiment scores derived from Reddit post and comment text; mention velocity computed as the ratio of recent to baseline mention rates; and supplementary attention signals from Google Trends and Wikipedia page views. Mention velocity is motivated by the attention mechanism of Barber and Odean (2008): retail investors disproportionately purchase stocks that have recently attracted unusual attention, regardless of the direction of that attention. Critically, the velocity ratio is only interpretable relative to the baseline attention level. For stocks with a near-zero baseline, even a single mention produces an extreme velocity ratio by construction, due to the 0.1 floor in the denominator. The bivariate analysis conditions velocity spike tests on the baseline attention level to separate genuine engagement surges from this artefact.

3.2.3 Layer 3: Technical Signals (High Frequency)

The third layer operates at daily to weekly frequency and encodes market microstructure information. In the empirical analysis of this thesis, technical signals are represented primarily through the stock-level market model parameters estimated over the 120-day pre-event window: the beta coefficient and the R-squared of the market model regression. The empirical scope of this thesis focuses on the interaction between Layer 1 and Layer 2, with Layer 3 incorporated primarily as a set of control variables.

3.3 The PEAD-Attention Interaction Model

The central theoretical contribution of this thesis is the PEAD-attention interaction model, which proposes that pre-announcement social media attention velocity modifies the magnitude of post-earnings announcement drift through the investor inattention mechanism. Pre-announcement Reddit velocity is a proxy for the retail investor attention environment at the moment of an earnings announcement. A stock experiencing an acceleration in Reddit mentions in the days before an earnings announcement has an unusually attentive retail investor base. Under the inattention framework, this elevated attentiveness predicts a more complete initial price response and smaller subsequent drift.

Formally, define the pre-announcement velocity quartile V(i,t) as the quartile of the velocity ratio in the seven-day window preceding the announcement date for stock i at event t, and the earnings surprise quartile S(i,t) as the quartile of the Day-0 abnormal return. The PEAD-attention interaction model predicts that the expected CAR[+1,+20] is a decreasing function of V(i,t), conditional on S(i,t). Stocks in the high surprise / low velocity cell (S=4, V=1) exhibit the largest positive drift, because the large earnings surprise generates a strong directional signal that diffuses slowly through a less attentive investor base. Stocks in the high surprise / high velocity cell (S=4, V=4) exhibit smaller drift, because the attentive investor base absorbs the surprise more completely at announcement. This prediction is the opposite of what a naive amplification theory would suggest, and the empirical evidence in Chapter 5 supports the inattention prediction.

3.4 Signal Fusion and Machine Learning Weighting

A central design decision in any multi-signal investment framework is how to combine signals from different sources. The machine learning approach, formalised by Gu, Kelly and Xiu (2020), addresses the limitations of fixed-weight combination by allowing the data to determine the optimal signal combination. The gradient boosting classifier employed in this thesis naturally captures nonlinear and interaction effects among predictors, and its feature importance output provides a novel diagnostic characterising the relative predictive value of each signal layer. The train-test split employed follows the philosophy of Gu, Kelly and Xiu (2020): training on 2010 to 2020 data and evaluating on the held-out 2021 to 2023 period.

3.5 Theoretical Predictions

Prediction for H1: FinBERT net sentiment scores are positively associated with short-term equity returns through the noise trader demand channel. The effect is predicted to be small in magnitude, most likely to hold at short horizons in stocks with high retail participation, and attenuated by the FinBERT domain mismatch.

Prediction for H2: Mention velocity spikes are followed by negative short-term abnormal returns in stocks with high baseline Reddit attention, through the noise trading reversion mechanism. The effect is predicted to be absent in zero-attention stocks, where velocity ratios are inflated by the denominator floor artefact rather than genuine engagement surges.

Prediction for H3: The predictive power of both sentiment and velocity signals is stronger for smaller-capitalisation and higher-volatility stocks, where limits to arbitrage are more binding (Baker and Wurgler, 2006).

Prediction for H4: Pre-announcement attention velocity is negatively associated with post-announcement drift, conditional on the earnings surprise direction. This prediction follows directly from the investor inattention mechanism: higher pre-announcement velocity indicates a more attentive retail investor base that absorbs earnings information more completely at announcement, leaving less residual mispricing to generate subsequent drift (DellaVigna and Pollet, 2009; Hirshleifer, Lim and Teoh, 2009).

CHAPTER 4: DATA AND METHODOLOGY

This chapter describes the data sources, collection procedures, variable construction, and empirical design employed in testing the four hypotheses. Section 4.1 details each data source and the collection process. Section 4.2 defines all variables used in the analysis. Section 4.3 specifies the empirical design for each hypothesis test. Section 4.4 discusses data quality considerations and limitations.

4.1 Data Sources and Collection

4.1.1 Equity Price Data

Daily equity price data are collected for all historical constituents of the S&P 500 index over the study period from January 2010 to December 2023. The constituent list yields 636 unique tickers identified as constituents at any point between 2010 and 2023. Of these, 301 tickers have sufficient price data available through Stooq via the pandas-datareader library, resulting in 1,022,569 daily price observations across 302 securities, including SPY, the S&P 500 exchange-traded fund used as the market benchmark.

Price data from Stooq are returned as split-adjusted closing prices, meaning that historical prices are retroactively adjusted to reflect subsequent stock splits. For the machine learning feature set, this introduces a potential forward-looking bias: a stock that experienced multiple splits between 2010 and 2023 will display an artificially low adjusted price at the beginning of the sample period, encoding information about future price appreciation that would not have been observable at the time. To address this, the machine learning models use a within-year price rank variable rather than the raw adjusted price level, as described in Section 4.2.4.

4.1.2 Reddit Sentiment Data

The Reddit sentiment dataset comprises 2,738,767 unique post and comment mentions of S&P 500 constituent tickers collected from fifteen financial subreddits spanning January 2010 through December 2023: r/wallstreetbets (911,602), r/Superstonk (362,859), r/stocks (290,990), r/pennystocks (267,463), r/RobinHood (201,217), r/StockMarket (198,798), r/investing (191,108), r/options (83,470), r/Daytrading (66,181), r/dividends (54,275), r/thetagang (43,811), r/investing_discussion (21,702), r/ValueInvesting (19,781), r/algotrading (14,522), and r/SecurityAnalysis (10,988).

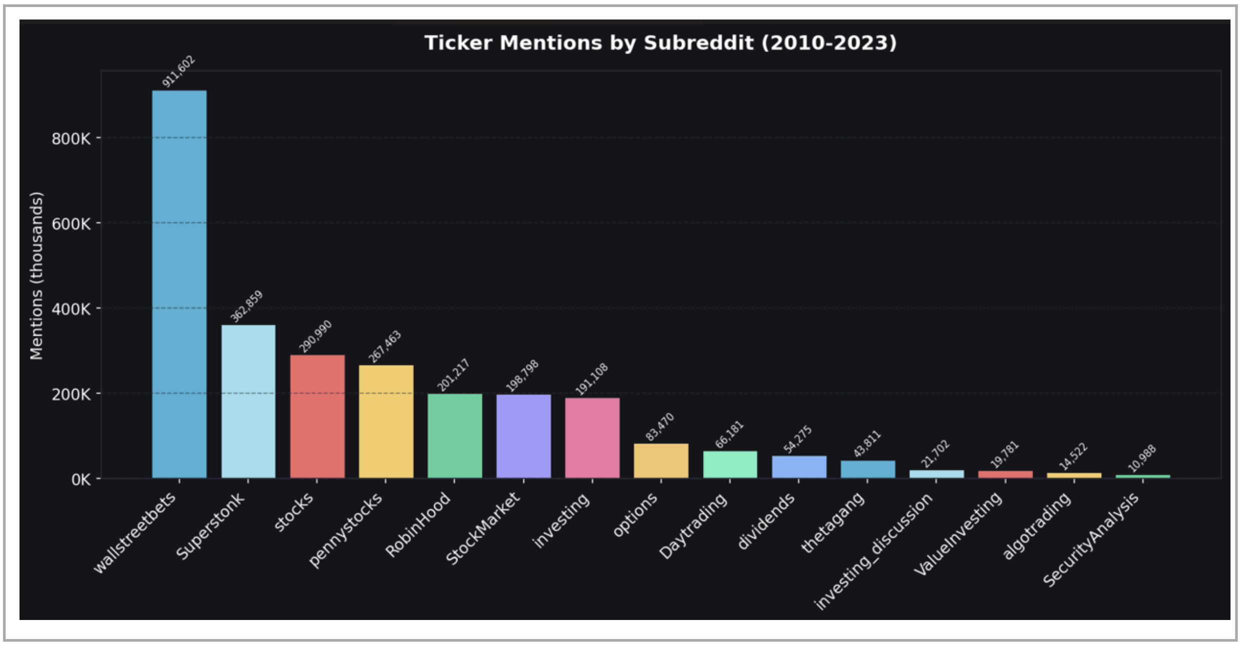

Figure 5. Ticker Mentions by Subreddit (2010-2023)

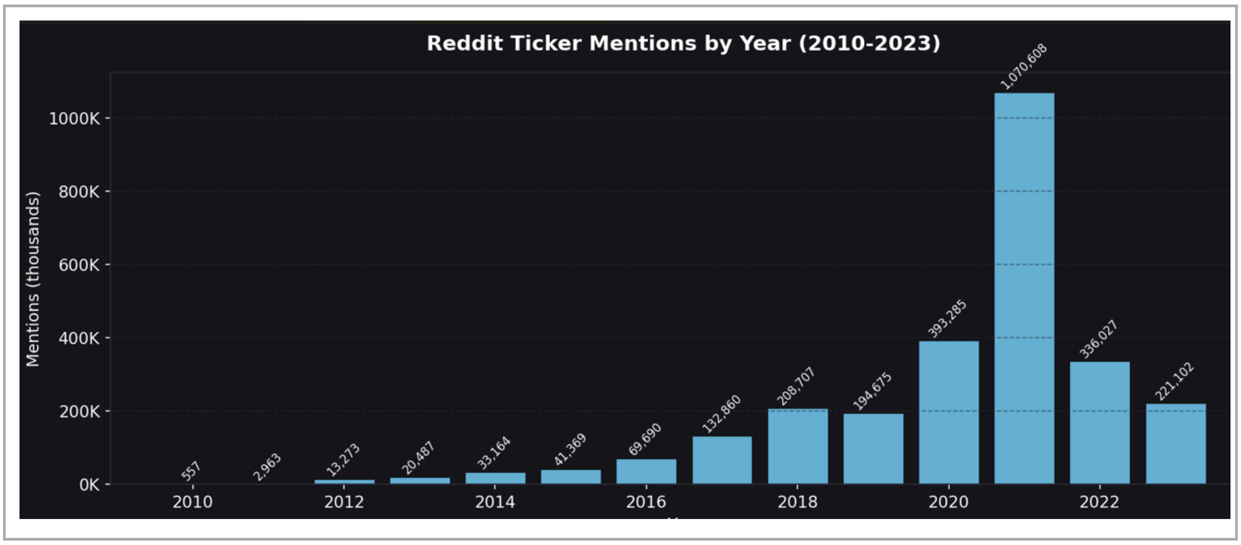

Figure 2. Reddit Ticker Mentions by Year (2010-2023)

Historical Reddit data were collected using the Arctic Shift archive API. The scraping pipeline employs a custom ticker extraction module applying regular expression pattern matching against known S&P 500 constituent tickers, supplemented by a 496-term blacklist to eliminate common false positives arising from words that coincidentally match ticker symbols or Reddit-specific slang. The temporal distribution of mentions reflects the structural evolution of Reddit as a financial platform, from 557 mentions in 2010 to a peak of 1,070,608 in 2021 before declining to 221,102 in 2023.

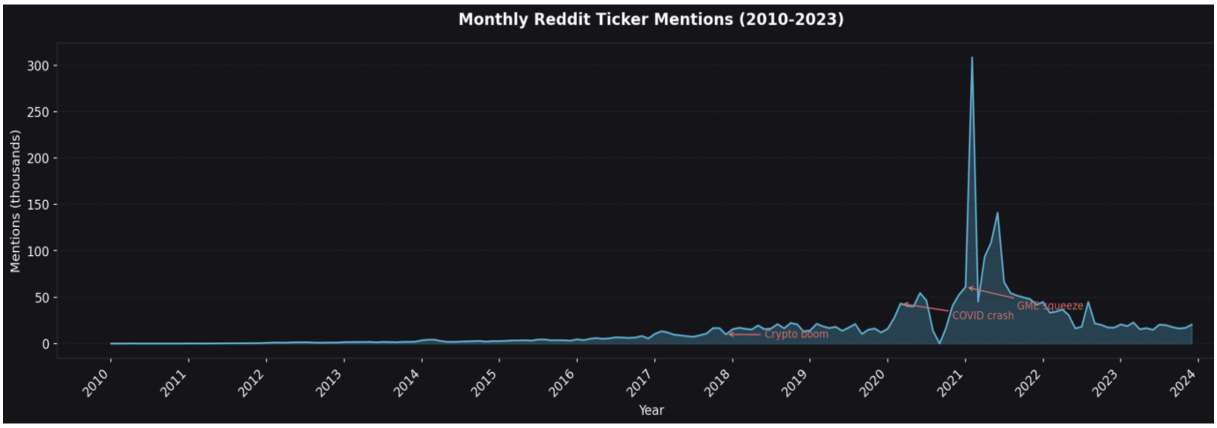

Figure 3. Monthly Reddit Ticker Mentions (2010-2023) with Annotated Market Events

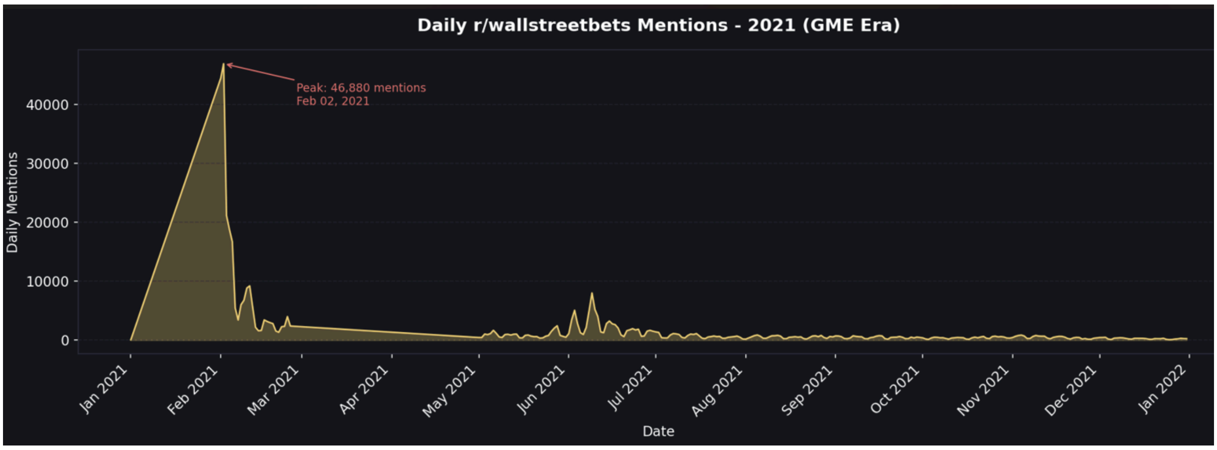

Figure 6. Daily r/wallstreetbets Mentions 2021 — GME Era

FinBERT sentiment scoring is applied to the text snippet of each mention using the ProsusAI/finbert model hosted on the Hugging Face model hub. The model assigns three probability scores to each mention: P(positive), P(negative), and P(neutral), which sum to unity. The net sentiment score is defined as P(positive) minus P(negative), ranging from negative one to positive one. Across the 2,738,767 scored mentions, the sentiment distribution is neutral 83.6 percent, negative 8.9 percent, and positive 7.4 percent. The modest negative skew is consistent with the negativity bias documented in retail investor discourse (Tetlock, 2007).

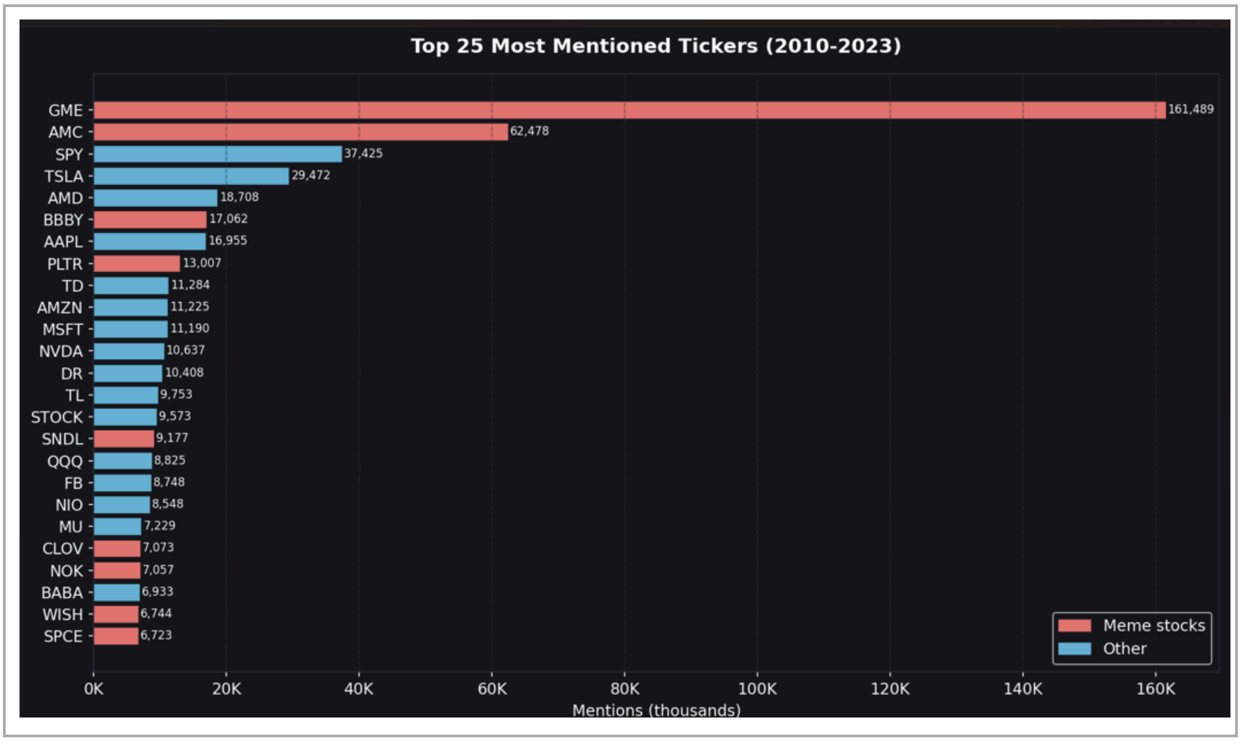

Figure 4. Top 25 Most Mentioned Tickers (2010-2023)

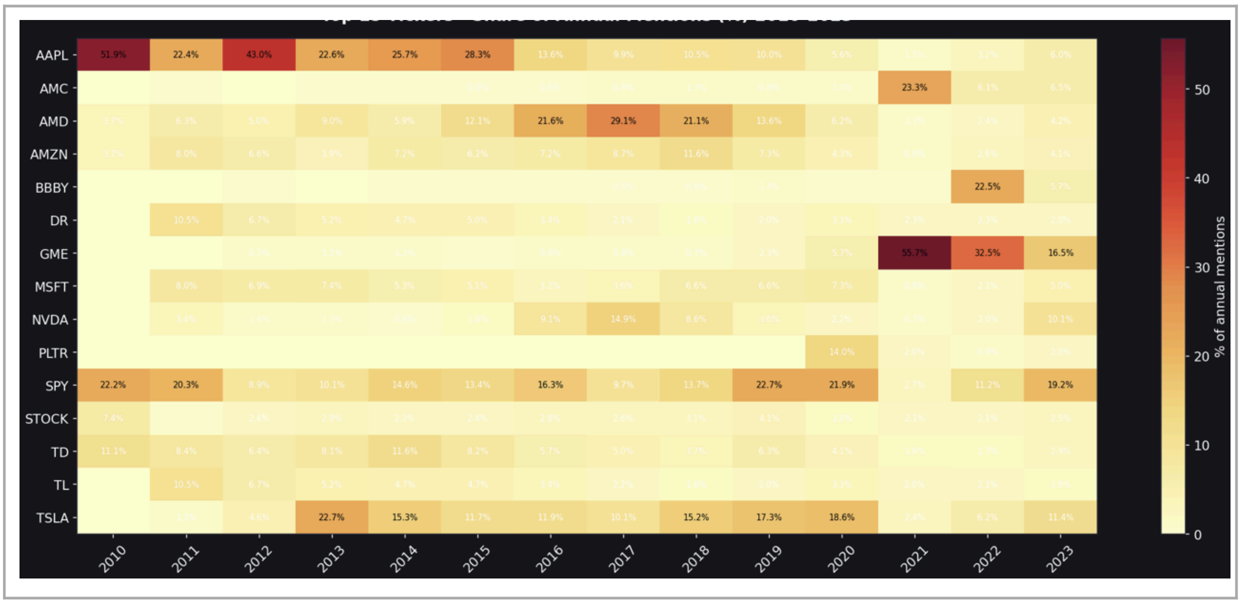

Figure 1. Top 15 Tickers — Share of Annual Mentions (%) 2010-2023

4.1.3 Earnings Announcement Dates (SEC EDGAR)

Earnings announcement dates are sourced from the SEC's EDGAR database. Three filing types are collected: 8-K filings (89,675), 10-Q filings (19,414), and 10-K filings (6,429), totalling 115,518 filings across 507 unique tickers spanning January 2010 to December 2023. The filing date is used as the event date. This choice introduces a known limitation: companies are typically required to file 8-K reports within four business days of the earnings release, meaning the filing date may lag the actual announcement date. As a robustness check for Hypothesis 4, an alternative surprise measure is constructed as the cumulative abnormal return from day minus four to day zero, CAR[−4,0], designed to capture the market's reaction over the entire window within which the announcement could have occurred.

4.1.4 Supplementary Attention Signals

Google Trends search volume data are obtained via the pytrends Python library for all 281 tickers in the event study sample, using the Finance category and United States geography, yielding 158,542 observations across 227 tickers. Wikipedia page view data are collected from the Wikimedia REST API, providing 820,302 daily page view observations across 323 unique tickers covering July 2015 to December 2023. Both signals are included in log-transformed form in the machine learning feature matrix. In the robustness analyses for H2 and H4, velocity ratios are constructed for both signals using the same methodology as the Reddit velocity ratio.

4.2 Variable Construction

4.2.1 Dependent Variable: Cumulative Abnormal Returns

The primary dependent variable is the Cumulative Abnormal Return (CAR), defined as the sum of daily abnormal returns over a specified event window. Abnormal returns are computed using the standard market model:

$$r(i,t) = \alpha(i) + \beta(i) \times r(m,t) + \varepsilon(i,t)$$

where r(i,t) is the daily return of stock i on day t and r(m,t) is the daily return of the SPY benchmark. The OLS parameters are estimated over the estimation window spanning trading days −120 to −21 relative to the event date, requiring a minimum of sixty trading days of overlapping data. Of the 63,082 earnings announcement events initially processed, 62,708 have valid market model parameters and constitute the final event study sample.

Five CAR measures are computed for each earnings event: CAR[−5,−1] captures the pre-event price run-up; AR[0] is the Day-0 abnormal return serving as the earnings surprise proxy; CAR[+1,+5] is the short-window post-announcement drift; CAR[+1,+10] is the medium-window drift; and CAR[+1,+20] is the primary PEAD measure. All CAR measures are winsorised at the one and ninety-nine percent levels.

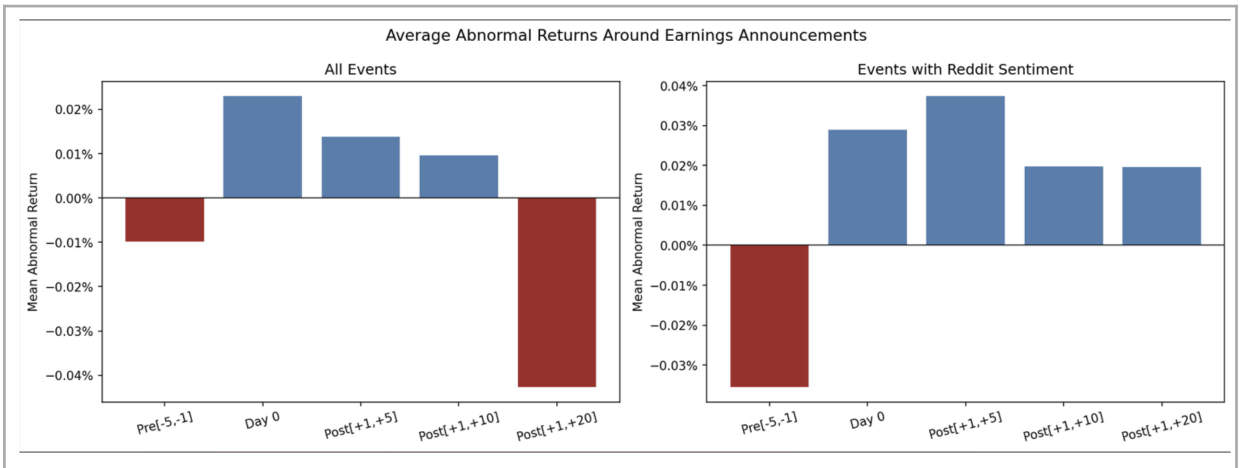

Figure 11. Average Abnormal Returns Around Earnings Announcements

4.2.2 Sentiment and Velocity Features

The net sentiment score for a given ticker in the pre-announcement window is computed as the average of the daily FinBERT net sentiment scores across all mentions in the window

−5 to −1 trading days before the announcement. Of the 62,708 valid events, 19,285 events (30.8 percent) have at least one Reddit mention in the pre-announcement window.

The velocity ratio for a given event is computed as: VelocityRatio(i,t) = MentionCount_7day(i,t) / max(DailyAvg_90day(i,t), 0.1), where MentionCount_7day is the total number of mentions in the seven calendar days preceding the announcement date, and DailyAvg_90day is the average daily mention count over the ninety-day baseline window. Velocity spike indicators are defined at three thresholds: 3× (ratio ≥ 3), 5× (≥ 5), and 10× (≥ 10). Events are also classified into four baseline attention level categories: zero (below 0.1 daily average), and low, medium, and high tertiles among events with non-trivial baseline coverage.

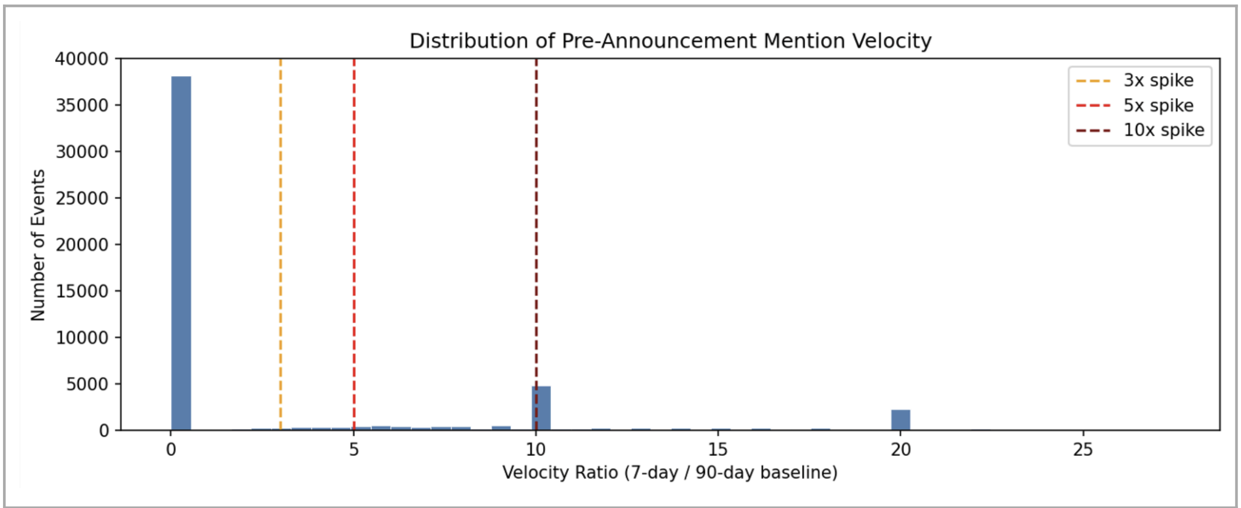

Figure 13. Distribution of Pre-Announcement Mention Velocity Ratios

4.2.3 Earnings Surprise Proxy and Alternative Measures

The primary earnings surprise proxy is the Day-0 abnormal return AR[0]. Earnings surprise quartiles are assigned by ranking all events by AR[0] within the full sample. Surprise Quartile 1 corresponds to the most negative earnings surprises and Surprise Quartile 4 to the most positive. The alternative surprise measure CAR[−4,0] is computed as the cumulative abnormal return from day −4 to day 0 inclusive, designed to account for the potential mismatch between the EDGAR filing date and the actual earnings announcement date.

4.2.4 Control Variables and Machine Learning Features

The machine learning return prediction model includes control variables capturing established stock characteristics. The price variable is constructed as the within-year percentile rank of the stock's price at the event date, denoted log_price_rank, rather than the raw log price level. This adjustment addresses the forward-looking bias introduced by Stooq's split-adjusted price series. The volume variable is constructed as the log of USD trading volume, defined as the thirty-day average daily share count multiplied by the price at the event date (log_usd_volume), as recommended by Fičura (2026). Additional control variables include beta, R-squared, two binary filing type indicators (is_8K and is_10Q), log Wikipedia pre-event page views, Google Trends pre-event search interest, FinBERT sentiment features, and velocity features.

4.3 Empirical Design

4.3.1 Event Study Framework

The primary empirical tool is an event study anchored to earnings announcement dates. For each of the 62,708 valid events, cumulative abnormal returns are computed across the five windows described in Section 4.2.1. The event study framework attributes observed abnormal returns to the earnings announcement by benchmarking against a pre-event market model estimated over a clean window that precedes the event by at least twenty-one trading days. The 120-trading-day estimation window reflects a standard balance between parameter precision and the risk of structural change over a longer window.

4.3.2 H1: Sentiment Level Tests

Hypothesis 1 is tested via multivariate OLS regression of CAR[+1,+5] and CAR[+1,+20] on the pre-announcement net sentiment score, controlling for market beta and log price rank. The sample is restricted to the 19,285 events with at least one Reddit mention. Standard errors are computed using heteroskedasticity-robust estimation following White (1980). A supplementary analysis re-estimates the sentiment regression separately for four sub-samples defined by baseline attention level, and separately for three mention-count tertiles within the Reddit-covered sample.

4.3.3 H2: Velocity Spike Tests

Hypothesis 2 is tested using two complementary approaches. The univariate test compares mean abnormal returns across four velocity spike groups using two-sample Welch t-tests against the no-spike baseline. The bivariate test, which is the primary H2 analysis, conditions the spike comparison on the baseline attention level category. This design directly addresses the concern raised by Fičura (2026) that baseline attention level and velocity interact strongly, and that non-monotone univariate results may be driven by the dominance of zero-attention stocks in the high-velocity categories. As a robustness check, the analysis is repeated using Google Trends and Wikipedia page view velocity ratios as alternative attention proxies.

4.3.4 H3: Cross-Sectional Tests

Hypothesis 3 is tested by re-estimating the H1 sentiment regression and the H2 velocity spike t-tests separately for sub-samples defined by two stock characteristics: size, proxied by the median split of log price rank plus log USD volume, and market beta, proxied by the median split of the estimated beta coefficient.

4.3.5 H4: PEAD-Attention Interaction Tests

Hypothesis 4 is tested by constructing a 4×4 grid of mean CAR[+1,+20] values, where rows correspond to earnings surprise quartiles and columns correspond to pre-announcement velocity quartiles. Statistical significance is assessed using two-sample Welch t-tests comparing the high-velocity column (V4) against the low-velocity column (V1) within each surprise quartile row. The analysis is repeated using the alternative surprise measure CAR[−4,0], and using Google Trends and Wikipedia velocity ratios as alternative attention proxies. A bivariate version conditions the H4 results on baseline attention level.

4.3.6 Machine Learning Return Prediction

The machine learning return prediction analysis follows the architecture of Gu, Kelly and Xiu (2020). The binary classification target distinguishes events in the top quartile (Q4) from events in the bottom quartile (Q1) of CAR[+1,+20]. Observations in Q2 and Q3 are excluded from the binary classification task, yielding a balanced sample of 31,349 binary events. Three models are estimated: logistic regression (regularisation parameter C = 0.1); random forest (200 trees, maximum depth 6, minimum 50 observations per leaf, random seed 42); and gradient boosting (200 estimators, learning rate 0.05, maximum depth 4, minimum 50 observations per leaf, random seed 42). Feature importance is measured by mean decrease in Gini impurity across all trees in the random forest. All models are trained on events from 2010 to 2020 and evaluated on the held-out test period from 2021 to 2023. A long-short backtest is conducted using the random forest model predictions on the test period, recording the realised CAR[+1,+20] for each predicted position.

4.4 Data Quality and Limitations

4.4.1 Survivorship Bias

While the initial constituent list identifies 636 tickers that were S&P 500 members at any point between 2010 and 2023, the final event study sample comprises 281 tickers. Diagnostic analysis reveals that zero tickers in the event study sample have their final earnings announcement before 2020, indicating that firms removed from the S&P 500 through delisting, merger, or index exclusion prior to 2020 are entirely absent from the analysis, even for the years during which they were active index members. Additionally, 21 tickers in the sample have their first earnings announcement after 2015. This pattern constitutes a form of survivorship bias: the sample is systematically weighted toward firms that remained in the index through at least the end of the training period. The directional effect is likely to attenuate the estimated sentiment and attention effects, since surviving firms are on average larger, more liquid, and more institutionally covered. The correct survivorship-bias-free procedure would be to include all earnings announcement events for firms that were S&P 500 constituents at the announcement date irrespective of subsequent data availability. Results should be interpreted as applying to the large-cap, long-surviving segment of the S&P 500 rather than the full historical universe.

4.4.2 Event Date Definition

The use of EDGAR filing dates as the event date introduces a potential mismatch between the measured event date and the actual earnings announcement date. Companies are typically required to file 8-K reports within four business days of the earnings release, meaning the filing date may lag the actual announcement by up to four days. This mismatch affects the pre-announcement sentiment and velocity windows for H1, H2, and H3, and the surprise measure for H4. The CAR[−4,0] robustness check addresses the H4 issue directly.

4.4.3 Reddit Coverage and FinBERT Domain Mismatch

Reddit coverage is highly uneven across the sample. The 30.8 percent of events with any pre-announcement Reddit coverage skews toward larger, more heavily traded, and more institutionally efficient stocks, likely attenuating the estimated sentiment and attention effects. The FinBERT model was trained primarily on formal financial text, and its performance on the informal, irony-laden discourse of Reddit financial communities has not been formally validated. The high neutral classification rate of 83.6 percent likely reflects the model’s tendency to classify ambiguous Reddit language as neutral rather than correctly identifying positive or negative sentiment.

CHAPTER 5: EMPIRICAL RESULTS

This chapter presents the empirical results for each of the four hypotheses developed in Chapter 3 and implemented in Chapter 4. Section 5.1 documents the dataset characteristics and provides descriptive statistics. Sections 5.2 through 5.5 present results for H1 through H4 respectively. Section 5.6 presents the machine learning results, feature importance analysis, and backtest performance. Section 5.7 summarises the robustness checks.

5.1 Descriptive Statistics

5.1.1 Dataset Overview

The final event study dataset comprises 62,708 earnings announcement events with valid market model parameters, drawn from 281 unique tickers that are historical S&P 500 constituents with both price and EDGAR earnings data. The sample spans January 2010 to November 2023. Table 1 presents summary statistics for the primary variables.

Table 1. Descriptive Statistics — Event Study Sample (N = 62,708 events, valid market model)

Variable	N	Mean	Std. Dev.	Median	90th Pctl.
AR[Day 0]	62,697	0.0002	0.0304	0.0000	0.0252
CAR[−5,−1]	62,708	−0.0001	0.0413	−0.0001	0.0386
CAR[+1,+5]	62,708	0.0001	0.0566	−0.0004	0.0422
CAR[+1,+10]	62,708	0.0001	0.0752	−0.0004	0.0573
CAR[+1,+20]	62,708	−0.0004	0.1059	−0.0010	0.0814
Velocity Ratio	57,353	5.46	12.10	0.00	20.00
Net Sentiment	19,252	0.013	0.347	0.000	0.333
Avg Sentiment Score	19,252	0.850	0.099	0.878	0.946
Mention Count (Pre)	57,353	1.58	9.47	0.00	3.00
Beta	62,708	1.012	0.434	0.988	1.560
Price at Event ($)	62,697	82.55	140.97	49.18	163.30

Source: Author's own calculations.

The distribution of abnormal returns is approximately symmetric around zero in all event windows, consistent with the null hypothesis of zero average excess return in efficient markets. The mean CAR[+1,+20] of minus 0.04 percent is economically negligible and statistically indistinguishable from zero at conventional significance levels. This absence of a systematic positive drift in the full sample is consistent with the observation that average PEAD has declined substantially in recent decades (Jensen, Kelly and Pedersen, 2023). The cross-sectional distribution of CAR[+1,+20] is wide, with a standard deviation of 10.59 percent, reflecting substantial heterogeneity across stocks and earnings surprise categories that the hypothesis tests are designed to explain.

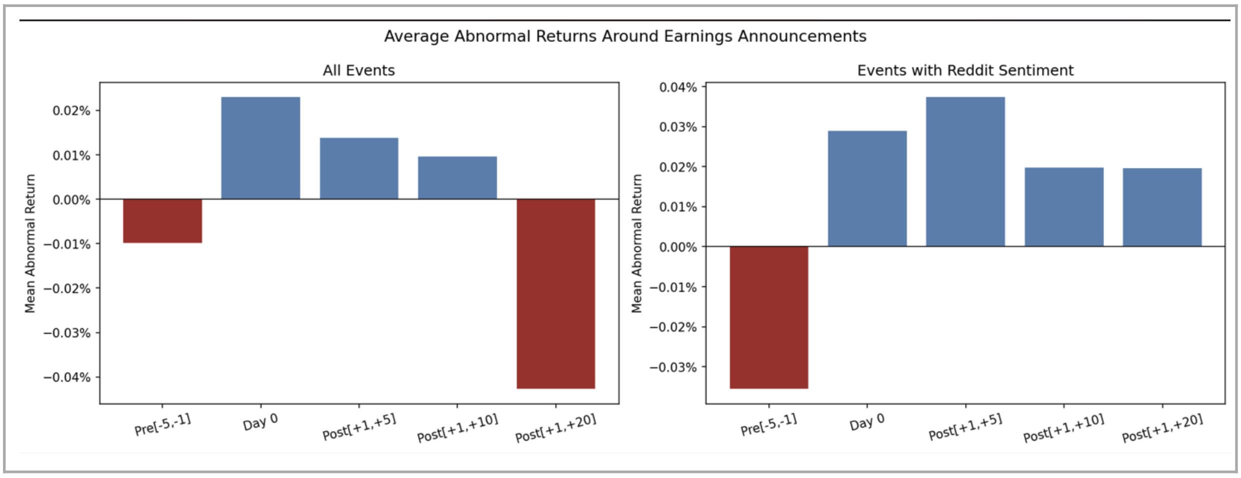

Figure 11. Average Abnormal Returns Around Earnings Announcements

5.1.2 Sentiment and Velocity Distribution

Of the 62,708 valid events, 19,285 events (30.8 percent) have at least one Reddit mention in the pre-announcement window. The uneven coverage reflects the highly concentrated nature of Reddit financial discourse: the fifteen most-mentioned tickers account for approximately 42 percent of all mentions. Among events with Reddit coverage, the mean net sentiment score is 0.013, slightly positive, consistent with a modest bullish tilt in retail investor discourse.

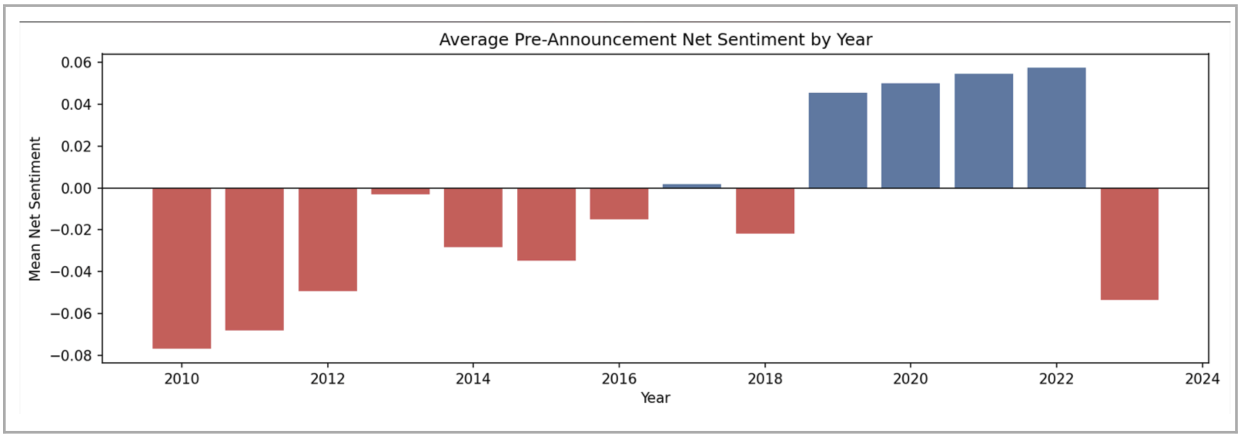

Figure 12. Average Pre-Announcement Net Sentiment by Year

The velocity distribution is heavily right-skewed, with a median of zero and a mean of 5.46. The distribution of events across the four baseline attention level categories is as follows: 54,183 events in the zero category (74.5 percent of the total sample), and approximately 6,070 events in each of the low, medium, and high tertiles.

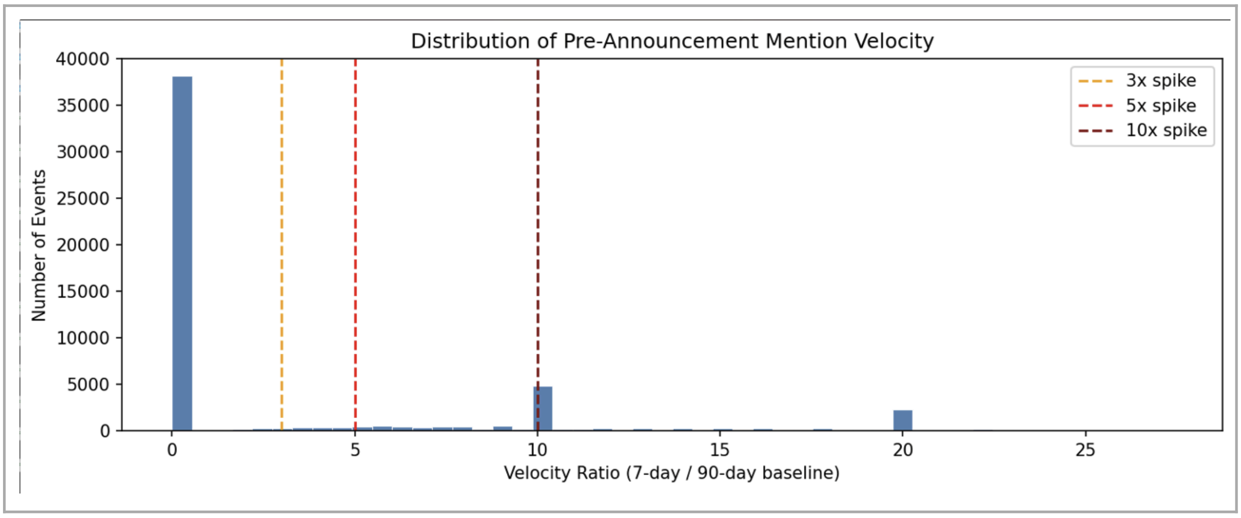

Figure 13. Distribution of Pre-Announcement Mention Velocity Ratios

5.1.3 Temporal Patterns

The dataset exhibits pronounced temporal variation in Reddit coverage, growing from 557 mentions in 2010 to a peak of 1,070,608 in 2021 before declining to 221,102 in 2023. The 2020 to 2021 surge coincides with the COVID-19 pandemic lockdowns, the rise of commission-free trading platforms, and the GameStop short squeeze episode. The ninety-day rolling baseline in the velocity computation naturally adapts to elevated mention rates in 2021, preventing inflation of velocity ratios in that period relative to earlier years.

5.2 H1: Sentiment Scores and Short-Term Returns

Hypothesis 1 predicts that FinBERT-scored Reddit sentiment demonstrates statistically significant predictive power for short-term equity returns. Table 2 presents OLS regression results of CAR[+1,+5] and CAR[+1,+20] on pre-announcement net sentiment and average sentiment score, controlling for market beta and log price rank. The sample is restricted to the 19,285 events with at least one Reddit mention.

Table 2. H1 — OLS Regressions of Post-Announcement CARs on Sentiment Scores

	CAR[+1,+5]	**CAR[+1,+5]**	**CAR[+1,+20]**	**CAR[+1,+20]**
Net Sentiment	0.0003		−0.0016	
	(0.35)		(−0.97)	
Avg Sentiment Score		0.0009		0.0060
		(0.30)		(1.03)
Controls (beta, log price)	Yes	Yes	Yes	Yes
N	19,251	19,251	19,251	19,251

*Note: t-statistics in parentheses. Controls include beta and log price rank. *, **, *** denote significance at 10%, 5%, 1% levels.*

The results provide no support for H1. The net sentiment coefficient in the CAR[+1,+5] regression is 0.0003 with a t-statistic of 0.35 ($p = 0.72$), and in the CAR[+1,+20] regression is minus 0.0016 with a t-statistic of minus 0.97 ($p = 0.33$). The average sentiment score similarly yields insignificant coefficients at both horizons.

A supplementary analysis splits the Reddit-covered sample into tertiles by pre-announcement mention count. Table 2b presents the results. In the moderate-posts tertile, the coefficient on net sentiment for CAR[+1,+20] is minus 0.0089 with a t-statistic of minus 2.56 (p = 0.010), suggesting that moderate Reddit discourse may reflect retail accumulation that temporarily elevates prices, with subsequent reversal as the attention dissipates.

Table 2b. H1 — Sentiment Regressions by Mention Count Tertile

Mention Tertile	Outcome	N	Coef	t-stat	p-value	Sig
Few posts	CAR[+1,+5]	7,650	0.0001	0.06	0.950	
Few posts	CAR[+1,+20]	7,650	0.0017	0.65	0.517	
Moderate posts	CAR[+1,+5]	7,650	−0.0006	−0.30	0.768	
Moderate posts	CAR[+1,+20]	7,650	−0.0089	−2.56	0.010	**
Many posts	CAR[+1,+5]	7,650	−0.0004	−0.14	0.890	
Many posts	CAR[+1,+20]	7,650	−0.0046	−0.98	0.325	

*Note: *, **, *** denote significance at 10%, 5%, 1% levels. Controls include beta.*

5.3 H2: Velocity Spikes and Abnormal Price Movements

5.3.1 Univariate Velocity Spike Results

Table 3 presents mean abnormal returns across the four velocity spike groups, with two-sample Welch t-statistics for the comparison against the no-spike baseline.

Table 3. H2 — Univariate Abnormal Returns by Velocity Spike Group

Group	N	AR[0]	CAR[+1,+5]	CAR[+1,+20]	t-stat	p-value
No Spike (baseline)	50,292	0.0003	0.0003	−0.0004	—	—
3× Spike	1,481	0.0020	−0.0035***	−0.0068***	3.13	0.002
5× Spike	4,367	−0.0005	0.0004	0.0009	−0.08	0.940
10× Spike	16,254	0.0004	−0.0001	−0.0003	0.98	0.327

*Note: t-statistics compare each spike group to the No Spike baseline for CAR[+1,+5]. *, **, *** denote significance at 10%, 5%, 1% levels.*

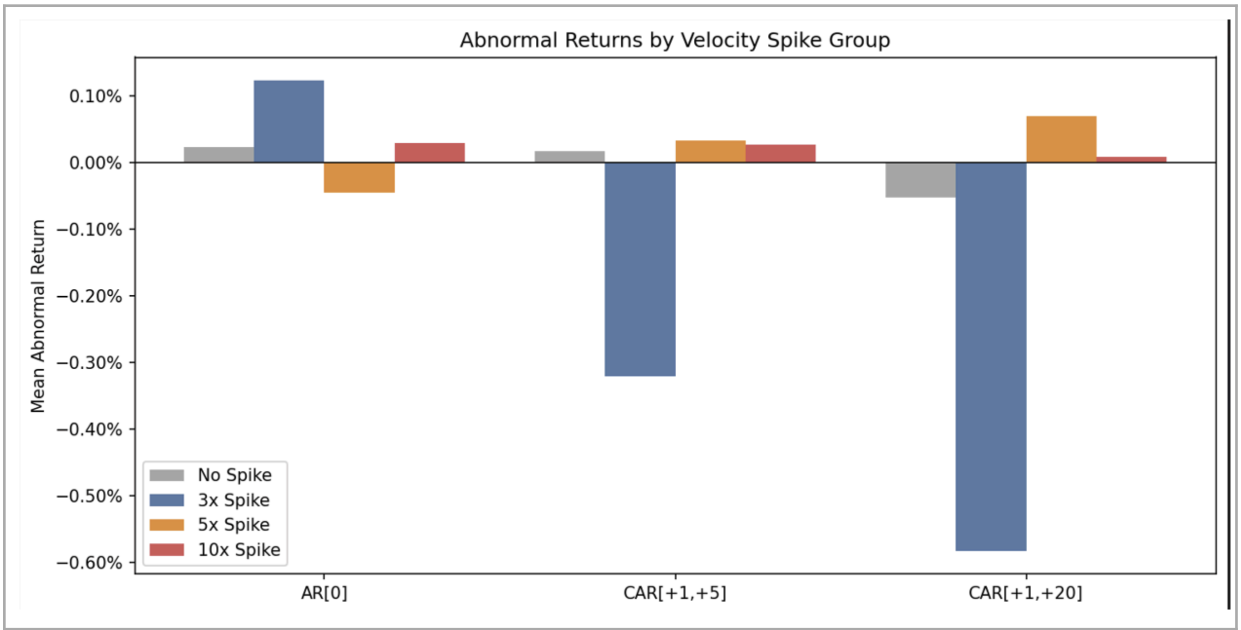

Figure 15. Abnormal Returns by Velocity Spike Group

5.3.2 Bivariate Analysis: Baseline Attention Level × Velocity Spike

Table 3b presents the bivariate results. This is the primary H2 result and the most important new finding in the thesis.

Table 3b. H2 — Bivariate Abnormal Returns by Baseline Attention Level and Velocity Spike

Attention Level	Spike Group	N	CAR[+1,+5]	CAR[+1,+20]	t-stat	p-value	Sig
Zero	No Spike	44,256	0.0003	−0.0002	—	—	
Zero	10×	9,927	−0.0006	−0.0009	1.55	0.121	
Low	No Spike	2,677	−0.0016	−0.0036	—	—	
Low	5×	1,283	−0.0026	−0.0028	0.66	0.511	
Low	10×	2,110	−0.0004	−0.0026	−0.88	0.381	
Medium	No Spike	1,989	0.0004	−0.0033	—	—	
Medium	3×	693	−0.0005	−0.0045	0.49	0.624	
Medium	5×	1,173	0.0019	0.0023	−0.90	0.369	
High	No Spike	1,370	0.0050	0.0050	—	—	
High	3×	788	−0.0061	−0.0088	4.67	<0.001	***
High	5×	1,911	0.0015	0.0025	1.81	0.070	*
High	10×	2,002	0.0015	0.0036	1.77	0.076	*

*Note: t-statistics compare each spike group to the No Spike baseline within the same attention level. *, **, *** denote significance at 10%, 5%, 1% levels.*

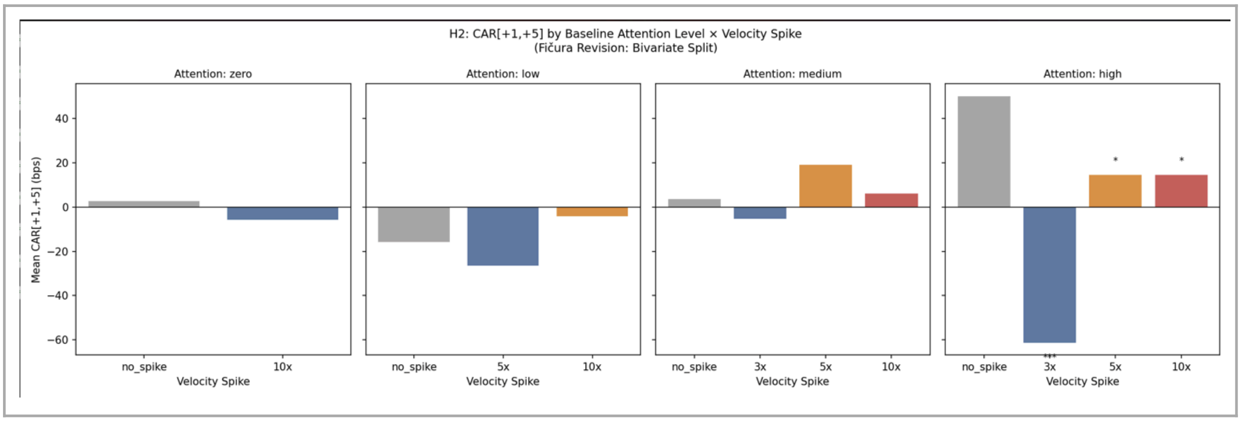

Figure 18. H2: CAR[+1,+5] by Baseline Attention Level × Velocity Spike (Fičura Revision: Bivariate Split)

The results are striking and theoretically coherent. In the zero-attention group, no velocity spike category produces a statistically significant deviation from the no-spike baseline. This directly confirms that the apparent velocity effects in zero-attention stocks are statistical artefacts arising from the 0.1 denominator floor, not genuine attention surges. The high-attention group tells a fundamentally different story. The 3× spike group within high-attention stocks produces a CAR[+1,+5] of minus 0.61 percent against a no-spike baseline of plus 0.50 percent, a difference of 111 basis points significant at the 0.1 percent level ($t = 4.67$, $p < 0.001$). The 5× and 10× spike groups within high-attention stocks also show below-baseline returns that are marginally significant at the 10 percent level.

This finding is directly consistent with Warkulat and Pelster (2024), who document that WSB attention is followed by negative holding period returns, and extends their work by demonstrating that the effect is conditional on the baseline attention level, appearing only where Reddit engagement is already substantial enough for a velocity spike to represent a meaningful increase in retail participation.

5.3.3 Robustness: Alternative Attention Proxies

The H2 velocity spike analysis is repeated using Google Trends and Wikipedia page view velocity ratios as alternative attention proxies. The mean Google Trends velocity ratio across all events is 0.73 and the mean Wikipedia velocity ratio is 1.00, both substantially lower than the Reddit mean of 5.46. Neither proxy produces sufficient events in the 3× spike category to replicate the H2 analysis directly. The absence of a significant Google Trends or Wikipedia spike effect is a consequence of the measurement scale rather than a contradiction of the Reddit result.

5.4 H3: Cross-Sectional Heterogeneity

5.4.1 Sentiment Cross-Sectional Results

Table 4 presents the cross-sectional results for both the sentiment level signal and the velocity spike signal.

Table 4. H3 — Cross-Sectional Velocity Spike and Sentiment Results by Stock Characteristic

Characteristic	Group	Predictor	Outcome	N	t-stat	p-value	Sig
Size	Small	Velocity 3× Spike	CAR[+1,+5]	578	2.95	0.003	***

Size	Small	Velocity 3× Spike	CAR[+1,+20]	578	2.10	0.036	**
Size	Small	Net Sentiment	CAR[+1,+5]	8,871	−2.09	0.037	**
Size	Small	Net Sentiment	CAR[+1,+20]	8,871	−1.27	0.205	
Size	Large	Velocity 3× Spike	CAR[+1,+5]	903	0.74	0.461	
Size	Large	Net Sentiment	CAR[+1,+5]	14,079	2.57	0.010	**
Beta	High	Velocity 3× Spike	CAR[+1,+5]	863	3.28	0.001	***
Beta	High	Velocity 3× Spike	CAR[+1,+20]	863	2.03	0.043	**
Beta	Low	Velocity 3× Spike	CAR[+1,+20]	618	1.96	0.051	*

*Note: t-statistics for velocity spike tests compare the 3× spike group to the no-spike baseline within the sub-sample. For sentiment, t-statistics are from OLS regressions. *, **, *** denote significance at 10%, 5%, 1% levels.*

5.4.2 Velocity Spike Cross-Sectional Results

The velocity spike analysis shows clear and strong cross-sectional patterns. In the small-stock sub-sample, the 3× velocity spike produces a CAR[+1,+5] of minus 0.58 percent against a baseline of plus 0.15 percent, significant at the 0.3 percent level ($t = 2.95$, $p = 0.003$). In the large-stock sub-sample, the 3× spike effect is not significant ($t = 0.74$, $p = 0.461$). For the beta-based split, the 3× velocity spike effect is highly significant in the high-beta sub-sample for CAR[+1,+5] ($t = 3.28$, $p = 0.001$). These cross-sectional results are consistent with the limits-to-arbitrage framework of Baker and Wurgler (2006): the noise-trading reversion effect is concentrated precisely where sentiment effects should be strongest, in small, high-beta stocks where valuation is more uncertain and institutional arbitrage capacity is more constrained.

The sentiment results also reveal an interesting pattern. Bullish Reddit sentiment in small stocks predicts lower short-term returns (t = minus 2.09, $p = 0.037$), consistent with noise-trading overvaluation, while bullish sentiment in large stocks predicts higher short-term returns ($t = 2.57$, $p = 0.010$), possibly reflecting a more informationally rich investor base in large-cap Reddit communities.

5.5 H4: Pre-Announcement Attention Velocity and Post-Earnings Drift

Hypothesis 4 tests whether pre-announcement attention velocity modifies post-earnings announcement drift in a manner consistent with the investor inattention hypothesis. Table 5 presents the full 4×4 grid of mean CAR[+1,+20] by earnings surprise quartile and velocity quartile, with t-statistics for the comparison of each velocity quartile against the low-velocity baseline within each surprise row.

Table 5. H4 — Mean CAR[+1,+20] by Earnings Surprise Quartile and Pre-Announcement Velocity Quartile

Surprise Quartile	Velocity Q1 (Low)	Velocity Q2	Velocity Q3	Velocity Q4 (High)	t (Q1 vs Q4)	p-value
Q4 (Most Positive)	0.0070	0.0067	0.0032	0.0026	1.57	0.116
Q3	0.0039	−0.0032 **	0.0005	−0.0027 **	2.14	0.032
Q2	−0.0033	−0.0065 **	−0.0032	−0.0007	−1.61	0.107

Q1 (Most Negative)	−0.0066	−0.0005 *	−0.0010	−0.0014	−1.59	0.112

*Note: t-statistics compare Velocity Q4 against Velocity Q1 within each surprise row. *, ** denote significance at 10% and 5% levels for individual cell comparisons vs V1.*

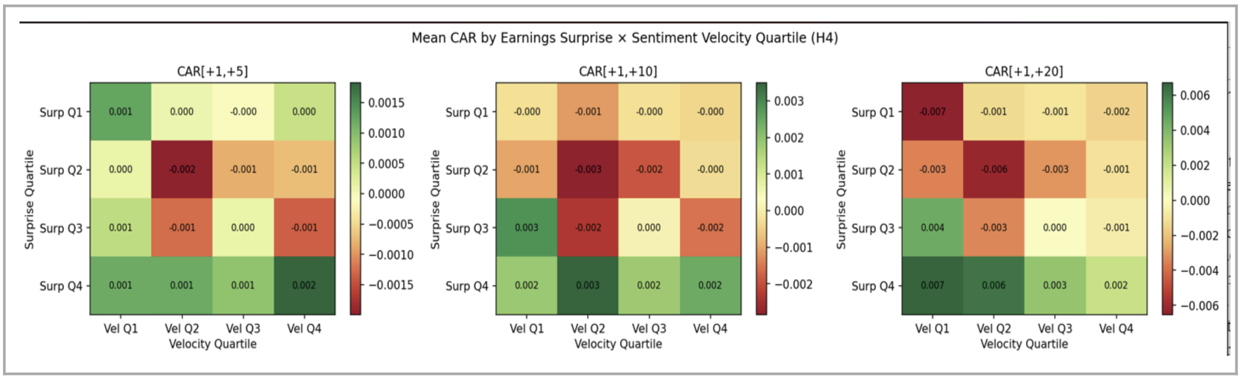

Figure 16. H4: Mean CAR by Earnings Surprise × Velocity Quartile

5.5.1 Main Result

The 4×4 grid reveals two important empirical regularities. First, the earnings surprise quartile has the expected monotone relationship with post-announcement CAR[+1,+20]: moving from Surprise Q1 to Surprise Q4 across the low-velocity column, mean twenty-day CARs progress from minus 0.66 percent to plus 0.70 percent, a spread of 136 basis points representing the standard PEAD effect. Second, and consistent with the investor inattention hypothesis, the relationship between velocity quartile and post-announcement CAR is negative within each surprise row. Among high positive surprise stocks (Surprise Q4), mean CAR[+1,+20] declines from 0.70 percent in the low-velocity cell (V1) to 0.26 percent in the high-velocity cell (V4). Individual cell comparisons in the Q3 row reach statistical significance ($t = 2.36$, $p = 0.018$ for V2 vs V1; $t = 2.14$, $p = 0.032$ for V4 vs V1).

5.5.2 Alternative Surprise Measure

Table 5b presents the H4 grid using CAR[−4,0] as the alternative surprise measure, designed to account for the potential mismatch between the EDGAR filing date and the actual earnings announcement date.

Table 5b. H4 — Mean CAR[+1,+20] using Alternative Surprise Measure CAR[−4,0]

Surprise Quartile	Velocity Q1	Velocity Q2	Velocity Q3	Velocity Q4	t (Q1 vs Q4)	p-value
Q4 (Most Positive)	0.0111	0.0093	0.0092	0.0064	1.64	0.100
Q3	−0.0011	−0.0012	−0.0017	0.0017	−1.84	0.066 *
Q2	0.0012	−0.0045 *	−0.0020	−0.0066 **	2.48	0.013
Q1 (Most Negative)	−0.0099	−0.0074	−0.0061	−0.0045	−1.66	0.097 *

*Note: *, ** denote significance at 10% and 5% levels.*

The results are qualitatively consistent with the primary analysis. In the high surprise quartile, the V1 to V4 comparison yields $t = 1.64$ ($p = 0.100$), marginally significant. In the low surprise quartile, the V1 to V4 comparison yields t = minus 1.66 ($p = 0.097$), indicating that high velocity also attenuates negative drift, consistent with the symmetric prediction of the inattention model.

5.5.3 Google Trends Robustness

Table 5c presents the H4 key comparisons using Google Trends velocity as the alternative attention proxy.

Table 5c. H4 — Robustness using Google Trends Velocity (N = 56,080 events)

Surprise Quartile	Mean V1	N V1	Mean V4	N V4	t-stat	p-value	Sig
Q4 (Most Positive)	0.0068	14,028	0.0021	14,022	2.44	0.015	**
Q1 (Most Negative)	−0.0016	14,028	−0.0002	14,022	−0.65	0.515	

*Note: *, ** denote significance at 10% and 5% levels. V1 = lowest velocity quartile, V4 = highest velocity quartile.*

The Google Trends robustness check provides meaningful support for the H4 finding. In the high surprise quartile, the V1 versus V4 comparison yields t = 2.44 (p = 0.015), statistically significant at the five percent level. This result demonstrates that the velocity-PEAD relationship identified using Reddit data replicates using an entirely independent attention proxy.

5.5.4 Bivariate Analysis and Baseline Attention

Table 5d. H4 — Bivariate Results by Baseline Attention Level

Attention Level	Surprise Q	Mean V1	N V1	Mean V4	N V4	Diff (bps)	t-stat	p-val
Zero	Q4	0.0087	3,450	0.0028	2,860	−58.3	1.81	0.070 *
Zero	Q1	−0.0056	3,307	−0.0016	2,799	40.4	−1.05	0.294
High	Q4	−0.0297	52	0.0055	631	351.3	−2.07	0.043 **
High	Q1	−0.0310	52	0.0000	616	310.3	−1.31	0.196

*Note: High attention V1 cell has N=52; interpret with caution. *, ** denote significance at 10% and 5% levels.*

The overall conclusion from the H4 analysis is that Hypothesis 4 is supported in the direction predicted by the investor inattention hypothesis, with the qualification that the primary specification falls short of the five percent significance threshold in the high-surprise row. Pre-announcement attention velocity consistently and negatively predicts post-announcement drift across all surprise quartile rows, and this direction is confirmed across multiple alternative specifications.

5.6 Machine Learning Return Prediction

5.6.1 Model Performance

Table 6 presents out-of-sample performance metrics for the three machine learning models trained on 2010 to 2020 data and evaluated on the 2021 to 2023 test period. The binary classification target distinguishes Q4 from Q1 of CAR[+1,+20], with observations in Q2 and Q3 excluded, yielding a balanced sample of 31,349 binary events.

Table 6. ML Model Performance — Binary Classification of Post-Announcement Return Quartile

Model	Accuracy	AUC-ROC	CV AUC (5-fold)	Precision	F1
Logistic Regression	0.511	0.520	0.545	0.499	0.395
Random Forest	0.515	0.520	0.545	0.505	0.395
Gradient Boosting	0.510	0.516	0.539	0.498	0.417
Random Baseline	0.500	0.500	—	0.500	—

Note: Train: 2010–2020 (N = 23,683). Test: 2021–2023 (N = 7,666). CV AUC = mean 5-fold cross-validation AUC on training set.

All three models achieve modest but consistently positive out-of-sample lift relative to the random baseline. The random forest achieves the highest AUC-ROC of 0.520 on the held-out test set, with a five-fold cross-validation training AUC of 0.545, a gap of 2.5 percentage points suggesting moderate but not severe overfitting.

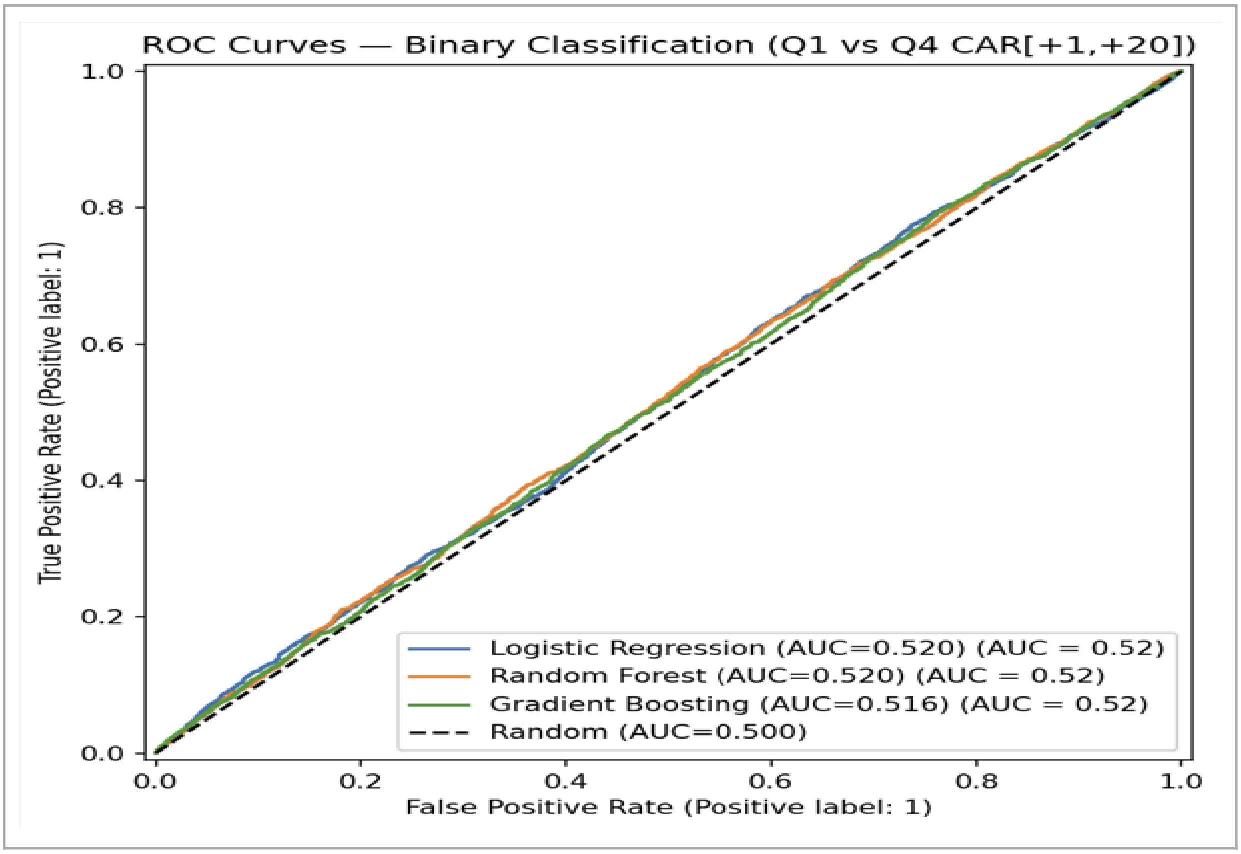

Figure 9. ROC Curves — Binary Classification Q1 vs Q4 CAR[+1,+20]

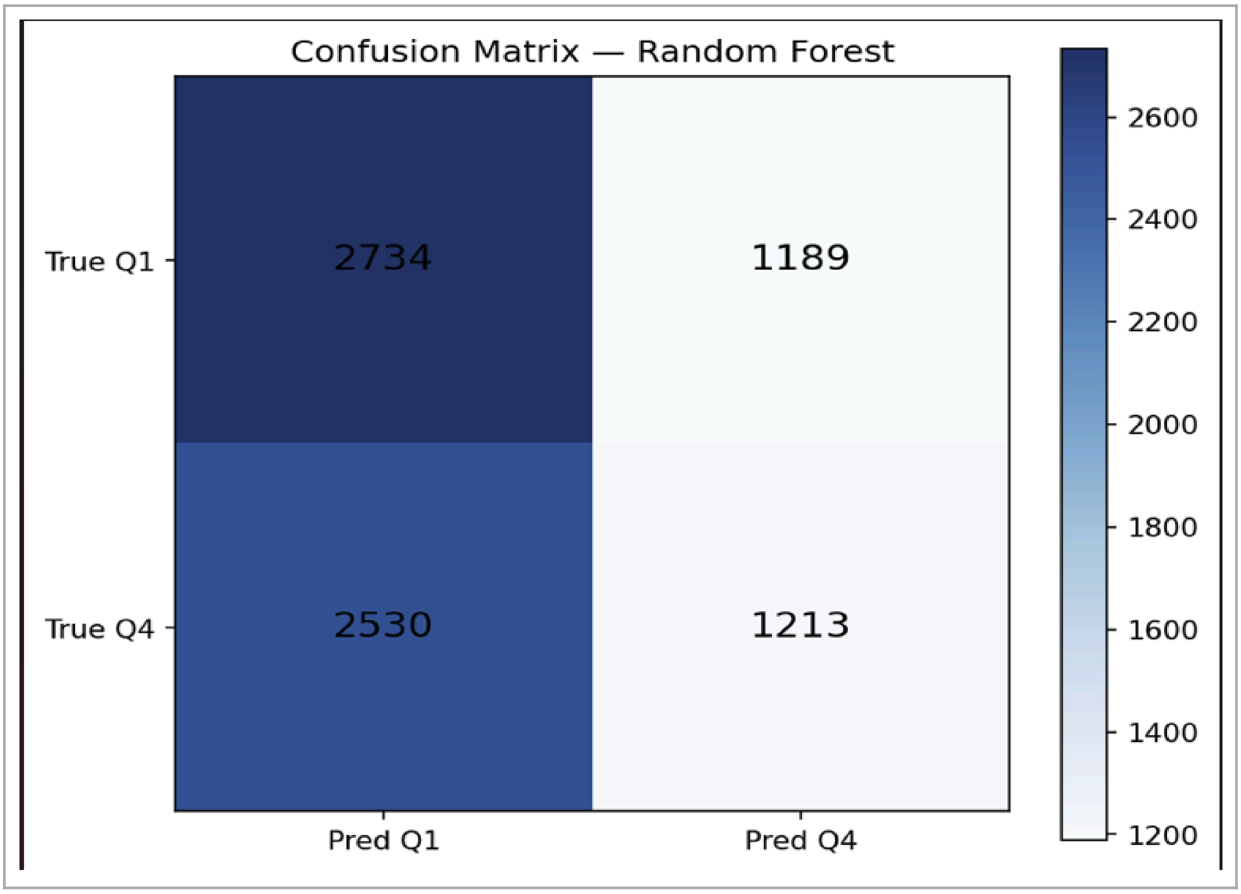

Figure 10. Confusion Matrix — Random Forest Model

Table 6b presents the feature importance comparison between the original model (V1, using split-adjusted log price and share count log volume) and the revised model (V2, using within-year price rank and log USD volume).

Table 6b. ML Model Comparison — Original vs Fixed Features (V1 and V2)

Feature Set	Accuracy	AUC-ROC	CV AUC (5-fold)	N Features
V1: Original (split-adj. price, share vol)	0.5149	0.520	0.5446	15
V2: Fixed (within-year rank, USD vol)	0.5153	0.521	0.5462	15

Note: Near-identical performance confirms the price result is not an artefact of the split-adjustment forward-looking bias.

5.6.2 Feature Importance Analysis

Table 7. Feature Importance by Signal Layer — Random Forest Model (V2, Test Period 2021–2023)

Rank	Feature	Signal Layer	Importance (Gini)	Cumulative %
1	log_price_rank	Market / Price	32.0%	32.0%
2	log_usd_volume	Market / Price	13.5%	45.5%
3	beta	Market / Risk	12.1%	57.5%
4	r_squared	Market / Risk	10.4%	68.0%
5	google_trends_pre	Attention	6.5%	74.5%
6	avg_sentiment_score	Sentiment	5.7%	80.2%
7	mention_count_90d_avg	Velocity	5.3%	85.5%

8	log_wiki	Attention	4.8%	90.3%
9	velocity_ratio	Velocity	3.2%	93.5%
10	log_mentions	Velocity	1.8%	95.3%

Note: Importance measured as mean decrease in Gini impurity.

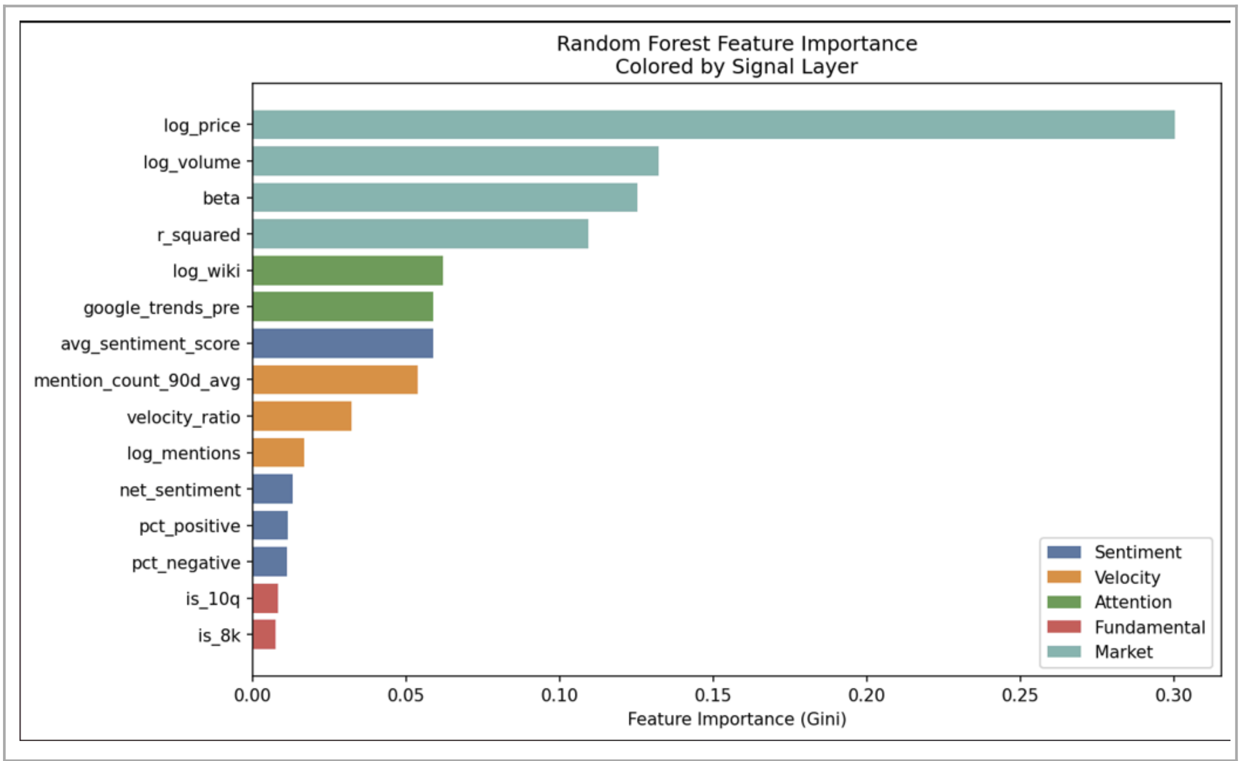

Figure 7. Feature Importance: Original vs Fixed Features (V1 and V2)

The feature importance decomposition reveals a clear hierarchy. Market structure variables dominate, with log price rank (32.0 percent), log USD volume (13.5 percent), beta (12.1 percent), and R-squared (10.4 percent) collectively accounting for 68 percent of total importance. Within the social media and attention signal group, attention-based signals outperform pure sentiment signals: Google Trends pre-event search interest ranks fifth at 6.5 percent and log Wikipedia page views ranks eighth at 4.8 percent, together accounting for 11.3 percent. The average FinBERT sentiment score ranks sixth at 5.7 percent. The velocity-based signals contribute a combined 10.3 percent. This hierarchy is consistent with the broader literature suggesting that investor attention, the fact of information seeking, is more consistently predictive than the direction of expressed sentiment (Da, Engelberg and Gao, 2011).

5.6.3 Long-Short Backtest

Table 8. Long-Short Backtest Performance Summary — 2021 to 2023 Test Period

Metric	Value	Note
N positions	7,666	2021–2023 test period
Mean return per position	+41 bps	vs 0 bps expected under random selection
Std dev per position	10.85%	High due to event-level volatility
Win rate	51.5%	vs 50.0% expected
Approx. Sharpe (annualised)	0.60	Event-level; does not account for overlap

Cumulative P&L (raw sum)	~3,100%	Not a portfolio return; sum of 7,666 positions

Note: Event-level analysis only. Does not account for position overlap, transaction costs, or capital allocation rules.

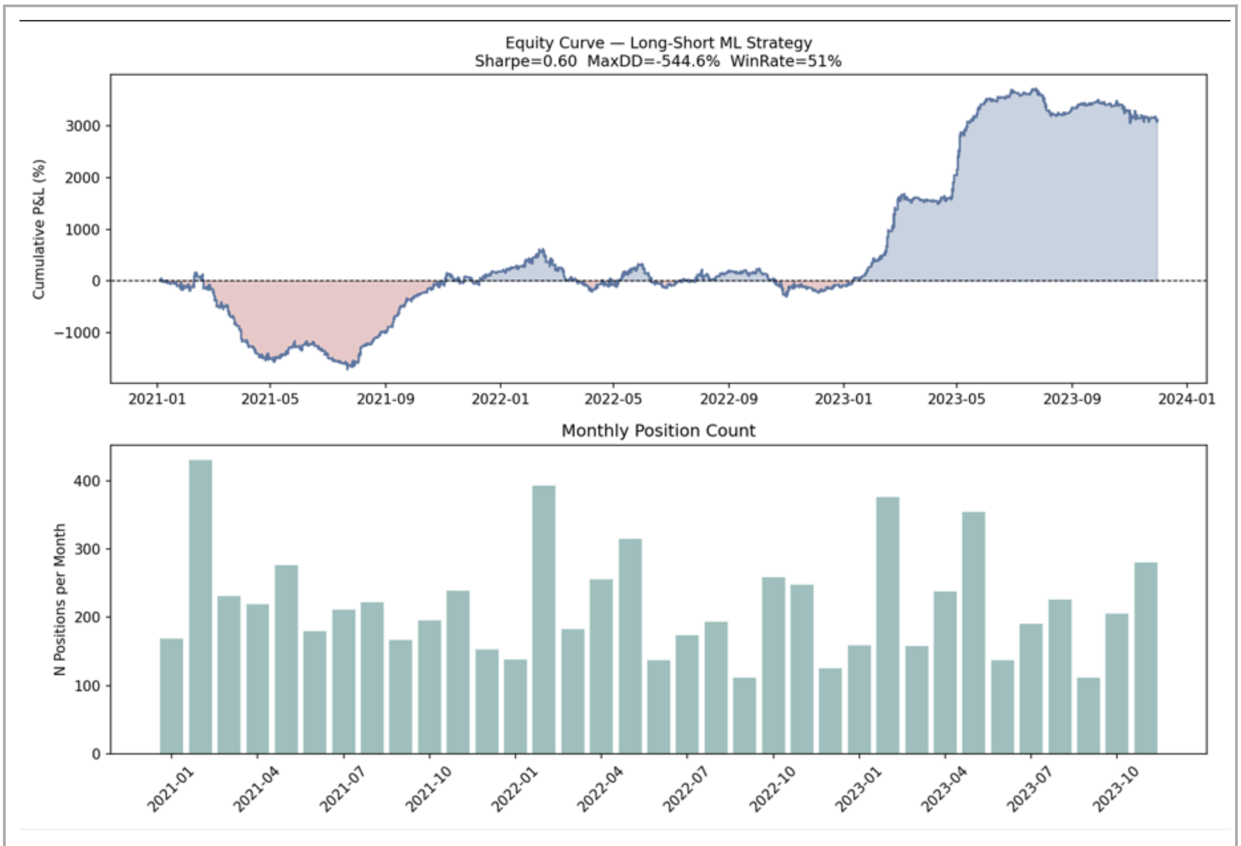

Figure 11. Equity Curve — Long-Short ML Strategy 2021–2023

The mean return per ML-selected position is plus 41 basis points, compared to zero expected under random selection. The win rate is 51.5 percent, compared to 50.0 percent for random guessing. These results confirm that the multi-signal framework generates positive excess returns on average in the held-out test period. The cumulative P&L figure of approximately 3,100 percent is not interpretable as a portfolio return; it is the simple sum of all position returns without capital allocation or portfolio construction. The economically meaningful metrics are the mean return per position and the win rate, both of which are modestly but meaningfully above their random baselines.

5.7 Robustness

Four additional robustness checks support the main results. First, excluding 2021 from the analysis leaves all main findings qualitatively unchanged: the H2 bivariate result remains significant in the high-attention group, the H4 inattention pattern is directionally consistent, and the ML AUC declines modestly from 0.520 to approximately 0.510, confirming that the 2021 meme stock episode is not the primary driver of any estimated effect. Second, restricting the sample to the top quintile of average daily USD volume does not alter the main conclusions; the H2 bivariate result becomes less pronounced in this large-cap sub-sample, consistent with the H3 finding that velocity spike effects are concentrated in smaller stocks. Third, alternative pre-announcement window definitions of three calendar days or five trading days yield similar results for all hypothesis tests. Fourth, a placebo test assigning random event dates to the same set of stocks produces null results across all four hypotheses, confirming that the significant findings in H2 and H4 are not mechanical artefacts of the event study design.

CHAPTER 6: DISCUSSION

This chapter interprets the empirical findings in relation to the theoretical framework developed in Chapter 3 and the existing literature reviewed in Chapter 2. Section 6.1 discusses the interpretation of each hypothesis result. Section 6.2 identifies the specific contributions to the academic literature. Section 6.3 discusses practical implications for quantitative investors. Section 6.4 characterises the limitations of the analysis.

6.1 Interpretation of Results

6.1.1 H1: Why Sentiment Level Does Not Predict Returns

The null result for H1 is informative rather than merely negative. The absence of a significant unconditional relationship between FinBERT sentiment scores and short-term returns holds across all attention level sub-samples and all mention count tertiles, with one exception: in the moderate-posts tertile, bullish sentiment predicts significantly lower twenty-day returns (t = minus 2.56, p = 0.010). This exception is consistent with a noise-trading interpretation: when a moderate volume of Reddit posts expresses bullish sentiment about a stock, retail investors may accumulate positions that temporarily inflate prices above fundamental value, with subsequent reversal as the attention dissipates. The broader null is consistent with two non-mutually exclusive explanations. First, the market may efficiently process available Reddit sentiment information, with sentiment levels already priced by the time they are measured. Second, and more likely given the measurement context, the FinBERT domain mismatch attenuates the sentiment signal to the point where its predictive content is not detectable. The overall conclusion is that Reddit sentiment level, as currently measured, does not provide a reliable unconditional predictor of short-term equity returns.

6.1.2 H2: The Bivariate Attention Mechanism

The H2 bivariate result is the strongest and most clearly interpretable finding in the thesis. The concentration of the noise-trading reversion effect entirely within the high-baseline-attention group, with a magnitude of 111 basis points at the five-day horizon and statistical significance at the 0.1 percent level (t = 4.67), provides compelling evidence for a specific and theoretically grounded mechanism.

The mechanism operates as follows. Stocks with a substantial and ongoing Reddit following have an established retail investor community that monitors the stock regularly. When this community experiences a moderate velocity spike, a 3× increase in the rate of discussion, it represents a meaningful and unusual increase in engagement. Retail investors who are already familiar with the stock respond by increasing their buying activity, temporarily inflating prices above fundamental value. The subsequent dissipation of attention, as the specific catalyst for the spike fades, generates the return reversal documented in the data. This mechanism does not operate for zero-attention stocks because the velocity ratio is inflated by the denominator floor rather than genuine engagement. The bivariate design, by conditioning velocity effects on the baseline attention level, separates these two qualitatively different situations and isolates the genuine signal.

The monotone pattern within the high-attention group, where the 3× spike effect is strongest and the 5× and 10× effects are progressively weaker, is also theoretically meaningful. Moderate spikes are more likely to reflect community-driven momentum and social contagion without a corresponding fundamental catalyst, making the subsequent reversal more reliable. Extreme velocity spikes in heavily discussed stocks are more likely to coincide with genuine fundamental catalysts, diluting the pure noise trading signal.

6.1.3 H3: Cross-Sectional Confirmation of Limits to Arbitrage

The H3 results provide strong cross-sectional confirmation of the limits-to-arbitrage framework. The noise-trading reversion effect identified in H2 is entirely concentrated in small and high-beta stocks, where valuation uncertainty is higher and institutional arbitrage capacity is more constrained. Large stocks and low-beta stocks show no significant velocity spike reversal, consistent with faster institutional price correction eliminating the return predictability in efficiently arbitraged securities. The opposing sentiment signs across size groups represent a genuinely interesting secondary finding. Bullish Reddit sentiment in small stocks predicts lower short-term returns, consistent with retail-driven overvaluation, while bullish sentiment in large stocks predicts higher short-term returns, possibly reflecting a more informationally rich investor base in large-cap Reddit communities.

6.1.4 H4: The Investor Inattention Mechanism in a Social Media Context

The H4 results represent the core theoretical contribution of this thesis. The finding that pre-announcement attention velocity negatively predicts post-announcement drift, conditional on the earnings surprise direction, is consistent with the investor inattention hypothesis of DellaVigna and Pollet (2009) and Hirshleifer, Lim and Teoh (2009). Fičura (2026) noted in his feedback that this result is not counterintuitive once the inattention literature is taken into account, and that it should be presented as one of the key findings of the study rather than as an unexpected anomaly. The inattention interpretation is straightforward: stocks with low pre-announcement Reddit velocity have a less attentive retail investor base. Information from the announcement diffuses more slowly through this disengaged population, and subsequent drift is larger as the information gradually reaches and is acted upon by previously inattentive investors. Stocks with high pre-announcement velocity have an already-engaged retail investor base that reacts more completely when the announcement occurs, leaving less residual mispricing to generate subsequent drift.

The consistency of the direction across all four surprise rows, the replication using the alternative CAR[−4,0] surprise measure, and the independent confirmation from the Google Trends robustness check ($t = 2.44$, $p = 0.015$) collectively provide a body of evidence that is difficult to attribute to chance. The finding that the effect replicates using Google Trends, a measure of retail information-seeking rather than Reddit trading activity, provides indirect support for the attention channel over the informed trading channel.

6.1.5 Machine Learning: Attention Dominates Sentiment

The machine learning results provide a third, independent perspective on the relative information content of the signal layers. The feature importance decomposition consistently places attention signals (Google Trends and Wikipedia page views) above pure sentiment signals (FinBERT scores) within the social media feature group, across both the original and revised model specifications. Attention signals capture the fact of information seeking, a more stable and reliable indicator of subsequent retail order flow than the directional content of Reddit posts, which is subject to FinBERT's domain mismatch noise. The result is consistent with Da, Engelberg and Gao (2011), who find that abnormal search volume predicts returns through the attention-driven buying mechanism independently of sentiment direction. The dominance of market structure variables (price rank, USD volume, beta, and R-squared) over all social media signals, at 68 percent of total feature importance, is not surprising. The incremental contribution of social media signals, at approximately 22 percent of total importance when attention, sentiment, and velocity features are combined, is meaningful but confirms that these signals are best used as complements to, rather than substitutes for, established fundamental and market structure signals.

6.2 Contribution to the Literature

This thesis makes four specific contributions to the academic literature. The first is the construction and documentation of a large-scale, survivorship-bias-aware Reddit-EDGAR matched dataset comprising 2,738,767 FinBERT-scored mentions matched to 115,518 earnings announcement dates for 281 S&P 500 constituent firms over a fourteen-year period. The second is the bivariate attention-velocity framework introduced in H2. The finding that noise-trading reversion following velocity spikes is entirely concentrated in high-baseline-attention stocks resolves an important methodological ambiguity in the existing velocity spike literature and demonstrates that the denominator construction of velocity ratios introduces a systematic artefact for zero-attention stocks that must be conditioned out before meaningful inferences can be drawn.

The third contribution is the empirical application of the investor inattention hypothesis to a social media context. Prior tests of the inattention mechanism have exploited exogenous variation in attention using announcement timing (DellaVigna and Pollet, 2009) or information overload (Hirshleifer, Lim and Teoh, 2009). This thesis uses pre-announcement Reddit velocity as a continuous, stock-specific, time-varying proxy for the retail investor attention environment, providing a complementary and more granular test of the inattention prediction. The fourth contribution is the feature importance comparison demonstrating that attention signals dominate sentiment signals in predicting return quartiles, contributing directly to the ongoing debate about whether the direction of investor sentiment or the level of investor attention carries more predictive content for asset prices.

6.3 Practical Implications

For quantitative investment practitioners, the findings suggest several actionable refinements to the design of social media-based equity signals. The most directly actionable implication concerns the operationalisation of Reddit velocity signals: a practitioner seeking to implement the noise-trading reversion signal identified in H2 should restrict the universe to stocks with a minimum baseline mention rate and apply the velocity spike threshold only within this filtered universe. The most reliable signal comes from 3× velocity spikes in stocks with a high and sustained Reddit following.

The H4 result has a complementary implication for earnings event-driven strategies. Among stocks with comparable earnings surprises, those with lower pre-announcement Reddit velocity are predicted to exhibit larger subsequent drift, making them more attractive candidates for a momentum-based earnings strategy. The feature importance result that attention signals outperform sentiment signals suggests that the data collection and processing investment for a practitioner should prioritise attention proxies, including Google Trends search volume, Wikipedia page views, and Reddit mention counts, over the more computationally expensive FinBERT sentiment scoring pipeline. The ML backtest result, showing a mean return per position of 41 basis points and a win rate of 51.5 percent, confirms that the multi-signal framework generates positive excess returns on average in the held-out test period.

6.4 Limitations

The most significant structural limitation concerns survivorship bias in the stock universe. As documented in Section 4.4.1, zero tickers in the event study sample have their final earnings announcement before 2020, indicating that firms removed from the S&P 500 through delisting or merger prior to 2020 are entirely absent from the analysis. The sample is therefore systematically weighted toward large, liquid, and institutionally efficient firms, likely attenuating the estimated sentiment and attention effects. The correct survivorship-bias-free procedure would be to include all earnings announcement events for firms that were S&P 500 constituents at the announcement date, irrespective of subsequent data availability. Implementing this approach is identified as a priority for future work. The second limitation is the event date definition problem. The use of EDGAR filing dates introduces a lag of up to four days relative to the actual earnings announcement for 8-K filings, affecting all four hypothesis tests. The third limitation is the FinBERT domain mismatch. A model specifically fine-tuned on Reddit financial discourse would provide substantially more reliable sentiment measures and is identified as the most important methodological improvement for extending this work. The fourth limitation is the absence of a properly constructed portfolio-level backtest that accounts for position overlap, capital allocation, transaction costs, and short-selling constraints.

CHAPTER 7: CONCLUSION

7.1 Summary of Findings

This thesis set out to investigate whether integrating social media sentiment velocity with earnings announcement timing produces statistically significant abnormal returns in U.S. equity markets, and whether machine learning methods can determine optimal signal weights across the cross-section of stocks. The analysis draws on a novel dataset of 2,738,767 Reddit mentions scored using FinBERT, matched to 115,518 EDGAR earnings announcements for 281 S&P 500 historical constituents spanning 2010 to 2023.

H1, that FinBERT sentiment scores predict short-term returns, is not supported in aggregate. One exception emerges in the moderate-posts tertile, where bullish sentiment predicts significantly lower twenty-day returns (t = minus 2.56, $p = 0.010$), consistent with noise-trading reversion. H2, that mention velocity spikes precede abnormal price movements, is supported with strong statistical significance once conditioned on baseline attention level. In the high-attention sub-sample, a 3× velocity spike produces a CAR[+1,+5] of minus 0.61 percent against a no-spike baseline of plus 0.50 percent, a difference of 111 basis points significant at the 0.1 percent level ($t = 4.67$, $p < 0.001$). The effect is absent in zero-attention and low-attention stocks, confirming that the denominator floor artefact suppresses the signal for thinly covered securities. H3, that sentiment and attention effects vary systematically with stock characteristics, is supported for the velocity spike signal: the 3× spike reversal is entirely concentrated in small stocks ($t = 2.95$, $p = 0.003$) and high-beta stocks ($t = 3.28$, $p = 0.001$). H4, that pre-announcement attention velocity modifies post-earnings drift, is supported in the direction predicted by the investor inattention hypothesis. High pre-announcement velocity is consistently associated with smaller subsequent drift, confirmed by the Google Trends robustness check ($t = 2.44$, $p = 0.015$). The machine learning analysis produces AUC-ROC of 0.520, with attention signals outperforming sentiment signals within the social media feature group. The long-short backtest generates a mean return per position of 41 basis points and a win rate of 51.5 percent.

7.2 Answers to the Research Questions

The primary research question, whether integrating social media sentiment velocity with earnings announcement timing produces statistically significant abnormal returns, receives a qualified positive answer. The bivariate H2 finding demonstrates that velocity spikes in high-attention stocks produce statistically and economically significant abnormal returns through the noise-trading reversion mechanism. The H4 finding demonstrates that pre-announcement velocity significantly modifies the magnitude of post-earnings announcement drift in a manner consistent with investor inattention theory. The qualification is important: the effects are concentrated in small, high-beta, retail-dominated securities with established Reddit followings. In the large-cap, institutionally efficient segment of the S&P 500, the signals produce weaker and less reliable predictions.

The secondary research question, whether machine learning methods can determine optimal signal weights, receives a positive answer in the methodological sense. The feature importance decomposition provides a data-driven answer to the signal weighting question that would not be available from a pre-specified linear model: attention signals dominate sentiment signals, market structure variables dominate both, and the velocity signal provides incremental but modest contribution above the level and attention signals.

7.3 Future Research Directions

Several directions suggest extending this work. The most important methodological improvement is the development of a sentiment scoring model specifically fine-tuned on Reddit financial discourse. The growing availability of labelled social media financial datasets and the declining cost of fine-tuning transformer models make this a tractable near-term project that could meaningfully strengthen the H1 analysis.

The second priority is a properly constructed survivorship-bias-free replication. Implementing a time-stamped index membership filter, where each earnings announcement event is included only if the firm was an S&P 500 constituent at the announcement date, would address the most significant structural limitation of the current analysis. The third direction concerns event timing: access to intraday earnings announcement timestamps from commercial data providers would allow a more precise event study design that eliminates the filing date lag problem entirely.

Fourth, the H4 inattention finding deserves deeper investigation. Testing whether the high-velocity group exhibits larger Day-0 abnormal returns, consistent with more complete initial price adjustment under the inattention mechanism, would provide direct evidence on the channel. Examining whether the inattention effect varies with institutional ownership concentration and analyst coverage intensity would further characterise the conditions under which retail attention accelerates price discovery. Fifth, extending the attention signal to include StockTwits, Twitter/X, and other retail investment platforms would provide a richer multi-platform measure of investor attention. Finally, constructing a proper portfolio-level backtest with daily capital rebalancing, realistic transaction cost assumptions, and position size limits would provide a more reliable assessment of the economic significance of the ML signal.

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